Predictive Planning with Workload-Responsive Correction for Port Gate–Yard Coordination

Abstract

Truck appointment systems and predict-then-optimize (PTO) planning can reduce port landside congestion when demand, compliance, and service-capacity models are reliable, but realized gate and yard workloads may diverge from the plan. We study execution-time workload correction in a coupled gate–yard system. At each booking cutoff, a fast rule adjusts only the quota for new appointment confirmations and yard-service capacity; confirmed appointments remain immutable. An optional paired-perturbation tracker moves the operating center. Conditional results explain why gate output propagates congestion downstream, why accurate open-loop PTO is a strong class-restricted benchmark, and why a maintained instruction creates cumulative yard drift when realized gate output persistently exceeds realized yard service. In public-data-calibrated vehicle-level experiments, high-fidelity fixed-pair PTO has the lowest cost and waiting. Under medium and severe model misspecification dominated by service-capacity overestimation, fixed-pair no-reidentification PTO develops persistent congestion that fast-plus-tracker feedback avoids; the archived design has no fast-only arm. A separate documented aggregate reconstruction compares fast-only and fast-plus-tracker variants with a cadence-triggered quota-recovery rule. It imposes the actions and activates the rule after a prespecified delay, rather than estimating capacity or re-solving PTO. Within this reconstruction, fast-only correction has lower diagnostic cost and terminal backlog than fast-plus-tracker feedback, while earlier activation limits post-warm-up persistence after an imposed overloaded pair. The evidence supports predictive planning as the backbone and workload correction as an execution safeguard when workload information arrives before the next planning revision.

Keywords: Port landside operations, Truck appointment systems, Gate–yard coordination, Feedback control, Predict-then-optimize, Model misspecification

1. Introduction

Port landside congestion is both an operational and a governance problem. Concentrated truck arrivals lengthen gate queues, make yard service uneven, and transmit delay to drayage operations and nearby road traffic (Guan and Liu, 2009; Torkjazi et al., 2018). Processing trucks faster at the gate is not sufficient: the yard must be able to absorb the workload released downstream. A gate policy that ignores yard pressure can therefore relocate congestion rather than remove it.

Truck appointment research has developed quotas, time-window designs, collaborative mechanisms, and robust rules for regulating terminal access (Guan and Liu, 2009; Torkjazi et al., 2018; Im et al., 2021; Azab et al., 2017; Li et al., 2024). Terminal-planning research likewise shows the value of integrated and robust resource decisions under uncertain demand and service (Rodrigues and Agra, 2022; Agra and Rodrigues, 2022; Zhen et al., 2022; Tan et al., 2017). Landside execution, however, is not a stationary replay of either plan. No-shows, late arrivals, exception trucks, road conditions, staffing, equipment availability, and yard disruptions separate the appointment–capacity plan from the workload actually observed at the gate and yard. The unresolved interface is how a terminal should respond between planning updates, after workload imbalance becomes visible but before the model is re-estimated and the plan is re-solved.

This paper asks when a container terminal should rely on predict-then-optimize (PTO) methods for appointment and capacity planning, and when execution-time workload correction should complement that plan. The conceptual framework distinguishes three operating regimes: executing a predictive plan whose model remains reliable, periodically re-estimating the model and re-solving the plan, and supplementing a maintained plan with workload-responsive correction during execution. The vehicle-level experiments compare maintained fixed-pair planning with workload correction; timely model maintenance is an interpretive boundary rather than an implemented same-error policy. The objective is to characterize these mechanisms, not to construct an automatic mode-selection system.

We develop a two-node gate–yard framework with an explicit decision clock. For each service window, the terminal issues an appointment quota and a yard-capacity instruction; the current request count then determines how many new requests are confirmed, while appointments confirmed for earlier cohorts remain immutable. Confirmed trucks can arrive on time, arrive late, or fail to appear, and gate completions become yard workload. The archived fast correction itself uses the gate queue and the yard queue plus busy yard units; a recent-request mean enters only the appointment-access floor applied after that correction. The policy does not observe the current request count or outstanding appointment pipeline when setting the quota. The previous quota affects the realized switching cost but is not a direct fast-feedback signal. A separable paired-perturbation tracker can move the operating center, but it is not needed to execute the fast correction.

The paper contributes four elements. First, it links appointment confirmation to downstream yard service while distinguishing pending requests, confirmed cohorts, and physical arrivals. Second, it specifies an interpretable workload-correction rule and separates its mandatory baseline, fast response, optional tracker, and safe execution map. Third, it develops conditional mechanism, model-fidelity, and simulation-selection bounds: gate output propagates to the yard, corrective value requires an informative and feasible response direction, and a maintained plan accumulates yard workload only when it induces a persistent realized gate–service imbalance. Fourth, it compares fixed operation, explore-then-select and Monte Carlo-selected fixed-pair PTO implementations, and feedback configurations in public-data-calibrated vehicle-level environments, supplemented by separately implemented diagnostics of component choice and response timing.

The remainder of the paper is organized as follows. Section 2 reviews related work on truck appointment systems, terminal planning under uncertainty, and queueing feedback control. Section 3 defines the gate–yard service setting and performance measures. Section 4 presents the workload-responsive feedback policy. Section 5 develops the analytical properties. Section 6 reports the public-data-calibrated experiments and separate mechanism diagnostics. Section 7 discusses managerial and policy implications. Section 8 concludes.

2. Literature review

2.1. Truck appointment systems, access planning, and compliance

Truck appointment systems are widely studied as tools for smoothing terminal arrivals and regulating access. Early work showed that quotas and time-window management can reduce peak gate demand (Guan and Liu, 2009). Later studies incorporated drayage tours, carrier cooperation, dynamic appointment design, robust decisions, and operational data (Torkjazi et al., 2018; Im et al., 2021; Azab et al., 2017; Li et al., 2024; Sun et al., 2025). This stream establishes appointment confirmation as a port-management decision and motivates the access outcomes reported here: waiting, confirmation, throughput, and unconfirmed requests.

Appointment decisions do not, however, determine the time at which confirmed trucks physically appear. No-shows, late arrivals, exception movements, fleet dispatch choices, and road disruptions separate a confirmed appointment cohort from the realized arrival stream. This timing distinction matters for both model design and policy evaluation: a rule that confirms few requests may report low queues without providing broad access, while a rule based only on current requests may overlook vehicles still arriving from earlier confirmed cohorts.

The appointment literature therefore supplies the upstream decision and the compliance mechanism. It does not by itself resolve how confirmed demand should be coordinated with the yard capacity needed to complete in-terminal service. That downstream planning problem is the focus of the next stream.

2.2. Predictive gate–yard planning and model maintenance

Yard and terminal-resource studies establish the downstream side of landside service. Yard templates, yard space, and integrated terminal planning show that internal capacity and resource allocation shape truck service performance (Tan et al., 2017; Zhen et al., 2022). Robust and distributionally robust terminal models protect plans against uncertain demand, vessel operations, or service capacity before execution (Rodrigues and Agra, 2022; Agra and Rodrigues, 2022; Li et al., 2024). These studies motivate the paper's gate–yard coupling: appointment confirmation controls upstream workload, whereas yard capacity determines whether gate completions can be absorbed downstream.

Predict-then-optimize uses an estimated outcome model to select operational decisions (Bertsimas and Kallus, 2020; Elmachtoub and Grigas, 2022). In a terminal, that logic appears when forecasts or estimated distributions determine appointment quotas and service resources before execution. Receding-horizon optimization goes further by updating the state or model estimate and repeatedly solving the planning problem (Mayne et al., 2000; Rawlings et al., 2017). The relevant distinction is therefore not simply offline versus online control. It is whether the predictive model remains accurate, whether the operator can re-identify it quickly, and whether queue observations are used between re-solving epochs.

This paper consequently treats PTO as a strong benchmark. An accurate open-loop model should perform well within its decision class, and an adaptive planner can recover from error when the relevant parameter is re-estimated and the planning problem is re-solved quickly. The no-reidentification case is narrower: it maintains an action selected under a belief model even after realized service departs from that belief. Separating these cases distinguishes the experimentally evaluated maintained-plan setting from the interpretation appropriate to adaptive planning.

The closest predictive studies thus provide the appointment–capacity plan, the uncertainty model, and the possibility of re-solving. The remaining interface is execution between planning updates: realized gate output becomes yard input, and current queues may reveal imbalance before a revised predictive model is available.

2.3. *Execution-time workload feedback and the remaining interface*

Queueing control provides a natural language for execution-time correction. Classical studies allow admission, service rates, or scheduling decisions to respond to congestion (Stidham and Weber, 1989; Wein, 1990; George and Harrison, 2001; Neely, 2010). Stochastic approximation and online queue-learning further motivate low-dimensional action updates from noisy performance observations (Robbins and Monro, 1951; Kushner and Yin, 2003; Borkar, 2008; Chen et al., 2026; Cohen et al., 2024).

Direct transfer to port landside operations is not immediate. The control concerns advance appointment cohorts rather than only contemporaneous admissions; confirmed trucks can arrive late; gate output becomes yard input; and yard capacity is discrete, shift-constrained, and disrupted. Appointment access must also be reported explicitly because low queues can be produced by confirming fewer requests. Finally, the relevant comparison is a PTO plan that may be difficult to improve upon when its maintained model is accurate.

The three streams therefore enter the paper in sequence. Appointment research defines the confirmed cohort and access consequences; predictive terminal planning links that cohort to downstream capacity and supplies the PTO benchmark; queueing control supplies an execution-time response to observed imbalance. This paper combines these elements to organize the distinction among maintained prediction, timely model maintenance, and workload correction in a coupled gate–yard system. Its vehicle-level experiments compare the maintained-plan and correction cases; a separate aggregate diagnostic examines recovery timing under a prespecified quota-recovery rule rather than implementing model re-identification.

The proposed correction layer is deliberately narrower than a terminal optimizer, yard-crane scheduler, or automatic mode-selection system. It adjusts two observable levers while detailed planning remains outside the layer. Table 1 summarizes how the three streams define the interface studied here.

3. System setting and operational decisions

This section translates the port-management problem into a sequence of operational events. For each service window, the policy first issues an appointment quota and activates yard capacity. The booking system then

Table 1: Literature streams and the operational interface studied in this paper.

Literature stream	Established focus	Interface addressed here
Truck appointment systems	Appointment quotas, time windows, carrier coordination, compliance, and terminal access	Distinguish pending requests, immutable confirmed cohorts, late arrivals, and access outcomes before linking confirmations to downstream service.
Predictive and robust terminal planning	Pre-execution appointment–capacity plans under uncertain demand and service; receding-horizon re-solving	Separate a maintained fixed-pair plan selected under a belief model from timely model maintenance and execution-time workload correction.
Queueing and online control	State-responsive admission, service, and low-dimensional stochastic updates	Convert observed gate–yard workload into feasible confirmation and yard-capacity corrections with integer and appointment-access constraints.

applies that quota to the realized current request count. The quota applies only to the new cohort; a request confirmed under it cannot subsequently be withdrawn by the feedback policy. Confirmed trucks may appear on time, appear late, or fail to appear; the gate processes realized arrivals; and each gate completion creates yard workload. We distinguish the issued quota q_t from the resulting confirmation count a_t throughout.

3.1. Port landside service ecosystem and decision timing

We consider the landside service ecosystem of a container terminal in which truck access decisions, gate processing, and yard operations are operationally coupled. Trucking companies submit requests for a specified service window before its booking cutoff. At that cutoff, the terminal confirms a subset of the still-pending requests. The confirmation decision applies only to the new cohort: appointments confirmed at earlier cutoffs remain valid and enter the outstanding pipeline until the truck appears or exhausts its lateness allowance. Confirmed trucks may arrive on time, arrive late, or fail to appear; trucks that complete gate processing subsequently require yard service. Existing truck-appointment studies show how time-window and confirmation decisions can regulate terminal arrivals (Guan and Liu, 2009; Torkjazi et al., 2018; Li et al., 2024). The terminal in our setting faces an additional coordination task: it must match newly confirmed upstream workload with sufficient downstream capacity.

The base model places the booking cutoff immediately before the associated service window. This convention lets the state at the cutoff inform both the new confirmations and the capacity activated for that window. At terminals that close appointment books earlier, the same decision applies to the next service cohort whose book remains open; it does not authorize cancellation of a closed cohort. Longer confirmation lead times would introduce an explicit action delay and are outside the present computational design. We therefore treat the truck appointment system as one control component within a broader port service ecosystem rather than as an isolated slot-allocation problem.

Figure 1 summarizes the decision sequence. Predictive planning sets the operating backbone before or between operating blocks, while the feedback layer uses realized gate and yard states to correct deviations during execution. The figure is a system schematic, not an empirical process map.

The operating horizon is divided into discrete decision windows indexed by

$$t = 1, \dots, T.$$

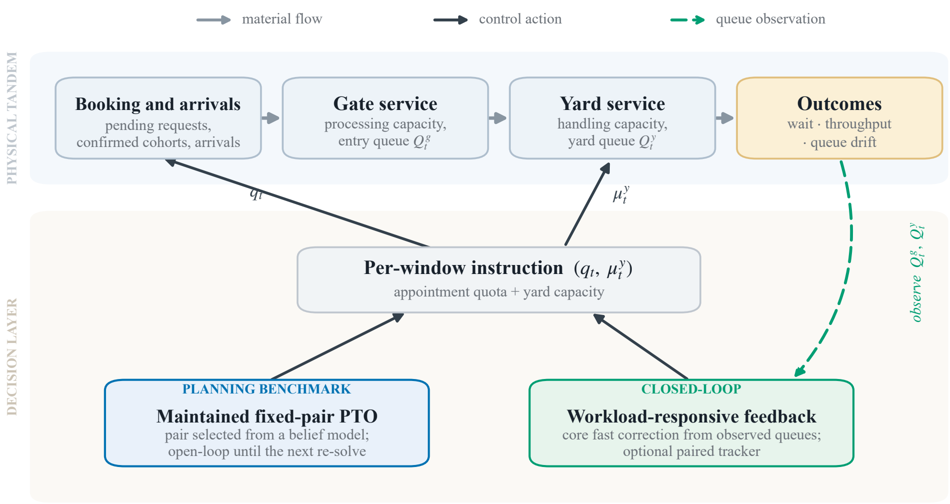

Figure 1: Schematic port landside service ecosystem. Both policy implementations set the same per-window instruction (appointment quota and yard capacity), but the maintained fixed-pair PTO benchmark remains open-loop between planning updates, whereas workload feedback observes realized gate–yard queues (dashed loop) and corrects imbalance subject to feasibility and an appointment-access floor. Confirmed appointments are not revoked.

The window length should be chosen for the operational response being modeled. The computational study uses 30-minute windows to resolve intraday queue responses; this is a simulation design choice rather than a resolution inferred from the daily and weekly public records. At the beginning of window t, the policy selects the instruction

$$\theta_t = (q_t, \mu_t^y),$$

where q_t is the integer appointment-quota instruction for the pending cohort and μ_t^y is an aggregate yard-service capacity level activated for that window. With R_t pending requests, the number newly confirmed is the mechanical outcome

$$a_t = \min\{R_t, q_t\}.$$

Neither q_t nor a_t cancels or reschedules a request already contained in the outstanding pipeline. The capacity variable may represent the number of active service units, a staffing–equipment package, or another implementable capacity tier. The two controls regulate newly confirmed workload and the capacity available to complete downstream service. We reserve *quota* for the issued instruction, *confirmation* for a_t, and *physical arrival* and *completion* for later events.

Let R_t denote the current appointment requests to which the issued quota is applied. Let L be the maximum number of windows by which a confirmed truck remains eligible to arrive late, and let $H_t = (H_{t,1}, \dots, H_{t,L})$ denote the simulator's appointment pipeline. Component $H_{t,d}$ records the still-eligible, not-yet-appeared requests confirmed for cohort $t - d$; thus d is cohort age at the current cutoff, not the number of lateness windows remaining. The base simulation sets $L = 2$; this is a finite-support design assumption, not a rule inferred from the public aggregate data. For a more general analytical model, C_t, O_t, M_t, and E_t can represent compliance information, container-yard occupancy excluding the truck service queue, equipment and service conditions, and exogenous disruptions.

The vehicle-level simulation also retains service already in progress. Let B_t^g and B_t^y be the gate- and yard-server status vectors at the cutoff, including the remaining service time of each busy server, and let I_t^g and I_t^y be the corresponding numbers of busy servers. A general analytical state is

$$\Xi_t = (Q_t^g, Q_t^y, B_t^g, B_t^y, H_t, C_t, O_t, M_t, E_t, \theta_{t-1}).$$

The archived simulator internally maintains the queues, server records, pending appearances, request history, and previous instruction needed to advance the event process. The general framework allows a controller to use a history $\mathcal{I}_t$, but the archived fast rule reads only

$$S_t^{\text{fast}} = (Q_t^g, Q_t^y, I_t^y, \overline{R}_t^{(6)}).$$

Here Q_t^g and Q_t^y are the numbers waiting at the gate and yard, I_t^y is the number of busy yard units, and $\overline{R}_t^{(6)}$ is the available mean over at most six preceding request counts, initialized at 25. The workload correction in Eqs. (11)–(16) uses only Q_t^g and $Q_t^y + I_t^y$; $\overline{R}_t^{(6)}$ enters only the appointment-access floor in Eq. (5). The current request count R_t, pipeline H_t, gate-busy count, recent-arrival summary, previous instruction, and current disruption realization are not direct workload-correction inputs. After the quota is issued, R_t truncates confirmations mechanically, the simulator advances H_t, and a disruption can reduce effective yard capacity. The previous quota enters the realized switching cost observed by the optional tracker. The event process evolves on the archived simulator state; S_t^{fast} is the complete input tuple for the executed fast policy, but it is not asserted to be Markov-sufficient for the event process.

All policies must satisfy the physical feasible set

$$\theta_t \in \Theta_t^{\mathrm{phys}},$$

with bounds

$$\underline{q}_t \leq q_t \leq \overline{q}_t, \qquad \underline{\mu}_t^y \leq \mu_t^y \leq \overline{\mu}_t^y.$$

The quota bounds define the smallest and largest instructions that the appointment system can issue; current request availability enters only through $a_t = \min\{R_t, q_t\}$, so confirmations cannot exceed pending requests. The physical set can also include shift or equipment availability constraints and limits on abrupt changes between adjacent windows. It contains no operation that removes an appointment from H_t. The feedback policy later adds an appointment-access constraint, producing a policy-specific subset $\Theta_t^{\mathrm{FB}} \subseteq \Theta_t^{\mathrm{phys}}$, where the superscript FB denotes feedback; the fixed and PTO benchmarks are evaluated within their stated candidate sets under the physical bounds. When yard capacity must be expressed as an integer number of active units, we write

$$m_t \in \{\underline{m}_t, \underline{m}_t + 1, \ldots, \overline{m}_t\}, \qquad \mu_t^y = h(m_t),$$

where $\underline{m}_t$ and $\overline{m}_t$ are the available lower and upper unit limits and $h(\cdot)$ maps equipment or staffing units into aggregate service capacity. The analytical model can use a continuous relaxation, while an implementation must map the relaxed instruction to a feasible integer level using a pre-specified deterministic tie-breaking rule.

The scope is a landside coordination layer. The model does not attempt to reproduce berth allocation, detailed container-stacking decisions, individual crane dispatch, external road-network routing, or microscopic truck trajectories. Its purpose is narrower: to examine how new appointment confirmation and yard-service capacity interact when realized workload deviates from a planned path.

3.2. Gate–yard tandem dynamics with appointment compliance

Appointment confirmation does not uniquely determine the number or timing of trucks that arrive. To make the timing explicit, let $X_{\tau,d}$ denote the number of requests confirmed for cohort τ that physically arrive $d \in \{0, \ldots, L\}$ windows later; the remaining confirmed requests in the cohort are no-shows. Let A_t^e collect walk-ins and other exception arrivals. The physical arrivals in window t are therefore

$$A_t = \sum_{d=0}^{L} X_{t-d,d} + A_t^e.$$

To make the appointment memory explicit, let $P_{\tau,t}$ be the number of requests confirmed for cohort τ that have not appeared by the cutoff of window t and remain eligible to arrive. Then $H_t = (P_{t-1,t}, \ldots, P_{t-L,t})$. After arrivals in window t, the new cohort enters the pipeline as $P_{t,t+1} = a_t - X_{t,0}$. For an earlier cohort with $1 \leq t - \tau < L$, $P_{\tau,t+1} = P_{\tau,t} - X_{\tau,t-\tau}$; after its final eligible window, any residual is removed as a no-show. With 30-minute windows, the base value $L = 2$ corresponds to a maximum modeled lateness of 60 minutes. The computational study separately checks one- and four-window supports because the aggregate public sources identify missed reservations but not the intraday distribution of lateness. Equivalently, the aggregate arrival kernel can be written as

$$A_t = A_t(a_t, H_t, C_t, E_t, \varepsilon_t),$$

where ε_t captures residual uncertainty. An equivalent conditional-moment representation is

$$\mathbb{E}[A_t \mid a_t, H_t, C_t, E_t] = \lambda_t(a_t, H_t, C_t, E_t),$$

and

$$\mathrm{Var}(A_t \mid a_t, H_t, C_t, E_t) = \sigma_t^2(a_t, H_t, C_t, E_t) > 0.$$

Thus, current arrivals depend on both the new confirmation decision and unresolved earlier cohorts, and they remain stochastic after those quantities are observed. The base simulation assigns confirmed trucks to on-time, one-window-late, two-window-late, or no-show outcomes and adds exception arrivals independently. This construction distinguishes pending requests, newly confirmed cohorts, immutable outstanding appointments, and physical arrivals.

Let $N_t^g = Q_t^g + I_t^g$ and $N_t^y = Q_t^y + I_t^y$ denote the total unfinished gate and yard workloads, including trucks waiting or in service. To distinguish potential service from completed service, let

$$G_t = G_t(B_t^g, M_t, E_t)$$

be the potential number of trucks that the gate can process during window t, before losses caused by within-window idleness or late arrivals are accounted for. Let D_t^g be the actual number of gate completions. In a vehicle-level system,

$$0 \leq D_t^g \leq \min\{N_t^g + A_t, G_t\}.$$

Similarly, let

$$Y_t(\mu_t^y) = Y_t(\mu_t^y, B_t^y, M_t, O_t, E_t)$$

be potential yard-service capacity under the chosen capacity level, and let D_t^y be actual yard completions. Then

$$0 \le D_t^y \le \min\{N_t^y + D_t^g, Y_t(\mu_t^y)\}.$$

The inequalities separate available work, potential capacity, and realized completion. They become equalities in an aggregate saturation model, or when work is available early enough to keep the relevant servers busy throughout the usable part of the window. In a vehicle-level discrete-event system, a truck arriving near the end of a window cannot recover capacity lost while a server was idle. Disruptions can further reduce potential yard capacity even when the planned capacity μ_t^y is unchanged.

The exact count balance for unfinished gate workload is

$$N_{t+1}^g = N_t^g + A_t - D_t^g. \tag{1}$$

The corresponding yard balance is

$$N_{t+1}^y = N_t^y + D_t^g - D_t^y. \tag{2}$$

Equations (1)–(2) are mass-balance identities and reveal the central operational coupling. The gate output D_t^g is the yard input. Increasing appointment confirmation can raise gate workload, while accelerating gate processing can shift trucks downstream more quickly. If realized yard service is not adjusted accordingly, an apparent improvement at the gate can coexist with persistent or increasing unfinished workload inside the terminal. Gate and yard performance must therefore be evaluated jointly.

The archived vehicle-level implementation is a first-come-first-served discrete-event simulator. Within each one-minute event step, newly appearing trucks join the gate queue, enter an available gate server in arrival order, move to the yard queue after gate completion, and then enter an available active yard server. Service that is unfinished at a decision-window boundary remains in progress, and gate- and yard-queue areas are accumulated minute by minute. The supplied reproduction archive contains the simulator source, experiment configurations, deterministic seed rules, and the raw replication outputs used for the three vehicle-level policy experiments and the high-fidelity component diagnostic. The documented independent aggregate reconstructions used for Tables 10 and 12 instead use aggregate queue and capacity approximations; their absolute values are not pooled with the vehicle-level model-fidelity results.

3.3. Performance measures, appointment-access protection, and capacity scope

The terminal balances congestion, resource use, operational adjustment, and appointment access. The archived vehicle-level implementation distinguishes the appointment-quota instruction from realized confirmations. Let $q_t \in \{12, \ldots, 30\}$ be the integer quota returned by the policy. The number of newly confirmed requests and the number left unconfirmed are

$$a_t = \min\{R_t, q_t\}, \qquad U_t^{\text{unc}} = R_t - a_t.$$

The cost uses the unconfirmed share

$$r_t^U = \frac{U_t^{\text{unc}}}{\max\{R_t, 1\}},$$

rather than the raw count U_t^{unc}. Let $\widetilde{m}_t$ denote the effective number of active yard units after the realized capacity disruption in window t, and let $\mathcal{Q}_t^g$ and $\mathcal{Q}_t^y$ be the time-average waiting-queue lengths accumulated minute by minute during the window. The implemented congestion-and-access score is

$$V_t = \mathbf{1}\left\{\mathcal{Q}_t^g > 20 \text{ or } \mathcal{Q}_t^y > 20\right\} + 5r_t^U.$$

It combines a congestion-threshold indicator with appointment-access loss; it is not a count of trucks that fail to complete service. Let ω_t collect the within-window appearance, service, exception-arrival, and disruption realizations. The dimensionless additive one-period cost implemented in the vehicle-level experiments is

$$\begin{aligned} c_t(\Xi_t, \theta_t, \omega_t) = w_g \frac{\mathcal{Q}_t^g}{20} + w_y \frac{\mathcal{Q}_t^y}{20} + w_r \frac{\widetilde{m}_t}{10} \\ + w_s \frac{|q_t - q_{t-1}|}{18} + w_v V_t. \end{aligned} \tag{3}$$

The divisors 20, 10, and 18 are, respectively, the queue reference, the maximum number of active yard units, and the appointment-quota span. The archived implementation has exactly these five weighted components. It penalizes quota changes but contains neither a separate yard-capacity-change penalty nor a container-yard-occupancy term. Legacy output files use the label "rejected requests" for U_t^{unc}. These cohort-specific requests exit the current decision; any later resubmission appears as a new request in a future $R_{t'}$. The variable does not include confirmed trucks that arrive late, remain in a queue, or fail to appear.

Let t_{w} be the number of warm-up windows excluded from evaluation. The expected post-warm-up average cost of a policy π under the true operating environment $\mathcal{M}$ is

$$J_{\mathcal{M}}(\pi) = \mathbb{E}_{\mathcal{M},\pi}\left[\frac{1}{T - t_{\mathrm{w}}}\sum_{t=t_{\mathrm{w}}+1}^{T} c_t(\Xi_t, \theta_t, \omega_t)\right]. \tag{4}$$

Let Π^{phys} be the class of nonanticipative policies satisfying $\theta_t = \pi_t(\mathcal{I}_t) \in \Theta_t^{\mathrm{phys}}$. The common decision problem underlying the policy comparisons is

$$\inf_{\pi\in\Pi^{\mathrm{phys}}} J_{\mathcal{M}}(\pi). \tag{4a}$$

The paper does not claim to solve Eq. (4a) over all admissible policies. The fixed baseline, predictive-planning policies, and workload-feedback policies studied below are restricted policy families with different information and updating structures. Writing the common problem explicitly ensures that their costs refer to the same gate–yard system and performance accounting.

Equation (3) is a normalized performance index rather than a financial or social-welfare measure. The computational study uses $t_{\mathrm{w}} = 7 \times 48$ windows and reports the post-warm-up average in Eq. (4), rather than a 30-day total. The archived source specifies all vehicle-level normalization maps, weights, and component calculations, so the reported vehicle-level weighted costs can be regenerated from the deposited configurations and raw outputs. Confirmation, throughput, unconfirmed requests, and resource hours per completed truck are also reported separately so that a low weighted cost cannot conceal reduced appointment access or additional resource use. The independent aggregate reconstructions use their separately documented diagnostic cost and are not pooled with the vehicle-level cost values.

The appointment-access safeguard can be imposed either as a penalty or as a constraint. In the computational feedback policy, the executed quota is subject to the rolling floor

$$q_t \geq \underline{q}_t^{\mathrm{acc}} := \max\left\{\underline{q}_t, \min\left\{\overline{q}_t, \lceil \rho_{\min}\overline{R}_t \rceil\right\}\right\} \quad\Longrightarrow\quad a_t \geq \min\{R_t, \underline{q}_t^{\mathrm{acc}}\}, \tag{5}$$

where $\rho_{\min} \in (0, 1]$ is a target confirmation fraction and $\overline{R}_t$ is computed from a pre-specified lookback W_R: $\overline{R}_t = |\mathcal{W}_t|^{-1}\sum_{s\in\mathcal{W}_t} R_s$, with $\mathcal{W}_t = \{\max(1, t - W_R), \dots, t-1\}$ and a pre-specified initial mean when $\mathcal{W}_t$ is empty. In the vehicle-level implementation, $W_R = 6$ decision windows; when no request history is yet available, the request mean is initialized at 25 requests per window. The nested bounds ensure that the quota floor remains physically feasible; current requests enter only afterward through $a_t = \min\{R_t, q_t\}$. We write Θ_t^{FB} for the subset of Θ_t^{phys} satisfying the quota floor in Eq. (5). This rolling floor is specific to feedback. The explore-then-select PTO surrogate separately filters out fixed quotas below 85% of its estimated mean requests, whereas the model-fidelity Monte Carlo candidate class has no access floor. Equation (5) protects the opportunity to receive an appointment, not physical arrival, service completion, or a waiting-time target.

Mean and tail waiting measures are retained as evaluation outcomes. For a waiting-time random variable W, the 95th percentile (P95) is

$$\mathrm{P95}(W) = \inf\{z : \Pr(W \leq z) \geq 0.95\}. \tag{6}$$

P95 waiting is not inserted as a primitive one-period cost because it is a path- or sample-level statistic rather than a naturally additive state cost. If tail congestion is to be optimized directly, an additive exceedance penalty such as $[Q_t - \kappa]_+$, or a formally defined conditional value-at-risk term, can be used. The computational study reports mean and P95 waiting, queue levels, throughput, confirmation, unconfirmed requests, utilization, and weighted performance cost.

Finally, we use a statewise capacity-slack condition to delimit an operating region in which corrective authority exists. Let $\mathcal{X}_0$ denote the analytical state region over which this condition is asserted, and write the physical feasible set as $\Theta_t^{\mathrm{phys}}(\xi)$ when it is state dependent. The condition is: there are constants $\delta_g, \delta_y > 0$ such that, for every $\xi \in \mathcal{X}_0$, there exists one action $\theta(\xi) = (q(\xi), \mu^y(\xi)) \in \Theta_t^{\mathrm{phys}}(\xi)$ that satisfies both

$$\mathbb{E}[A_t \mid \Xi_t = \xi, \theta_t = \theta(\xi)] \leq \mathbb{E}[G_t \mid \Xi_t = \xi] - \delta_g, \tag{7}$$

and

$$\mathbb{E}[D_t^g \mid \Xi_t = \xi, \theta_t = \theta(\xi)] \leq \mathbb{E}[Y_t(\mu^y(\xi)) \mid \Xi_t = \xi, \theta_t = \theta(\xi)] - \delta_y. \tag{8}$$

The quantifier order is therefore $\forall \xi \in \mathcal{X}_0 \, \exists \theta(\xi)$, not the stronger requirement that one fixed action work for every state. Equations (7)–(8) define the region in which a corrective action is physically available; the drift results below still require the evaluated policy to choose an action that actually improves the realized input–service imbalance. The condition is not derived from the feedback rule and is neither used as, nor sufficient for, a stochastic-stability theorem. Persistent failure for every feasible action indicates structural capacity shortage rather than an algorithmic failure.

4. Workload-responsive feedback coordination

The coordination mechanism separates three operations that a reader must not conflate: choosing a physical baseline, applying a per-window fast correction, and optionally tracking a moving operating center. The baseline may come from an offline predictive plan, a documented operating rule, or another independently chosen feasible pair. The fast rule then moves the appointment quota and yard-capacity instruction when observed workload reveals an imbalance. The optional paired-perturbation tracker changes the center from sequential realized costs; it is not required to execute workload correction. The primary vehicle-level high-fidelity and model-misspecification comparisons evaluate the combined fast-plus-tracker implementation. A high-fidelity component comparison and an independent aggregate severe-error reconstruction separately evaluate the tracker's incremental effect.

4.1. Policy families and information structure

We organize the evaluated policies into three families that represent different uses of operational information. A policy first issues an appointment–capacity instruction and then passes it through a physical execution map Γ_t^{phys}, which clips the quota and capacity to their available levels and applies pre-specified integer rounding. Current request availability then produces $a_t = \min\{R_t, q_t\}$. This map does not impose the feedback-specific access floor. The non-responsive family is represented by a fixed instruction

$$\theta_t = \Gamma_t^{\mathrm{phys}}(\vartheta^F), \qquad \vartheta^F = (q^F, \mu^{y,F}), \qquad t = 1, \ldots, T, \tag{9}$$

or, more generally, a predetermined intraday instruction pattern that is not revised in response to realized queues. The realized confirmation count can still fall below q^F when fewer requests are available; this mechanical mapping is not state feedback.

The predictive-planning family is anchored by predict-then-optimize (PTO), which serves as the strong planning benchmark. PTO follows the predictive-to-prescriptive logic of selecting a decision through an estimated model of uncertain outcomes (Bertsimas and Kallus, 2020; Elmachtoub and Grigas, 2022). A planning implementation selects an appointment–capacity plan from an outcome model and then executes the resulting instruction without queue feedback. In the model-fidelity stress test, the selected pair is maintained throughout the evaluation horizon without re-estimating capacity from realized completions. We call this experimental benchmark *fixed-pair no-reidentification PTO*. Let $\vartheta_t = (q_t, \mu_t^y)$ denote a planned appointment quota and yard-capacity instruction, and let

$$u = (\vartheta_1, \dots, \vartheta_T) \in \mathcal{U}$$

be an open-loop appointment-capacity plan. Physical bounds induce the instruction map

$$\chi(\vartheta_t) = \Gamma_t^{\mathrm{phys}}(\vartheta_t) = \theta_t.$$

This mechanical execution makes the issued instruction physically feasible without using gate or yard queues and without imposing the feedback-specific access floor. The state variable R_t then determines the realized confirmation count. PTO uses a belief model $\widehat{\mathcal{M}}$ for demand, compliance, service times, and capacity, and solves

$$u^{\mathrm{PTO}} \in \arg\min_{u \in \mathcal{U}} \widehat{J}(u), \qquad \widehat{J}(u) = \mathbb{E}_{\widehat{\mathcal{M}}}\left[\frac{1}{T - t_{\mathrm{w}}} \sum_{t=t_{\mathrm{w}}+1}^{T} c_t\big(\Xi_t, \chi(\vartheta_t), \omega_t\big)\right]. \tag{10}$$

The general plan class in Eq. (10) may vary across the day, but it is open-loop because execution is not revised using realized gate and yard queues. The computational study uses the narrower finite subclass

$$\mathcal{U}_{\mathrm{pair}} = \big\{u_\vartheta = (\vartheta, \dots, \vartheta) : \vartheta \in \Theta^{\mathrm{cand}}\big\} \subseteq \mathcal{U},$$

where $\vartheta = (q, \mu^y)$ specifies one intended appointment quota and yard-capacity level and Θ^{cand} is a finite subset of the physical instruction set. The vehicle-level study uses two distinct selectors within this class. The high-fidelity comparison, access study, and component study use an explore-then-select implementation: during the first 20% of the 30-day run, the quota spans 12–30 and the yard level cycles through 6–10; the resulting request and appearance summaries parameterize a deterministic queueing surrogate that scores the eligible subset of the 95-pair grid after filtering quotas below 85% of estimated mean requests. The selected pair is then maintained unless a rolling-surrogate variant is enabled. The six-day exploration lies inside the seven-day warm-up, so the reported metrics exclude it, although the simulated state carries through to evaluation. The component run also contains a 24-hour update variant using a trailing 144-window record and a three-hour update variant using a trailing 48-window record. These variants refresh request and appearance summaries and rerank the surrogate; they do not re-identify gate or yard service capacity.

The model-fidelity experiment instead evaluates every pair by four Monte Carlo planning paths under each fidelity-specific belief model, selects the lowest estimated-cost pair before policy evaluation, and then maintains that pair. Equation (28a) applies to this selector, not to the explore-then-select or rolling-surrogate implementations. The separate aggregate recovery diagnostic introduced in Section 6.4 uses neither selector: it imposes an overloaded pair, an always-on fixed-pair reference, and a state-dependent quota-recovery rule.

The workload-feedback family is anchored by fast workload-responsive correction. It does not reconstruct the full joint distribution of arrivals, compliance, and service before acting. Instead, it shifts the appointment quota and yard capacity when observed queues indicate an accumulated flow imbalance. Its instruction is passed through a feedback execution map Γ_t^{FB}, which applies the physical map and then enforces Eq. (5), so the resulting instruction belongs to Θ_t^{FB}. This policy uses more execution-time information than fixed-pair no-reidentification PTO because it observes realized queue states, and it operates under an additional access constraint. It therefore requires observable workload signals, meaningful feedback signs, feasible corrective authority, and sufficient capacity for corrective actions to matter. The paired-perturbation tracker introduced later changes the operating center but is not part of the core definition.

This information structure permits a documented plan or operating rule to supply the physical baseline while queue feedback corrects departures from it. In the computational study, the vehicle-level predictive-planning variants isolate model fidelity within the implemented fixed-pair candidate class. The separate aggregate diagnostic examines the delay before a prespecified action-revision rule becomes available; it does not represent model estimation or re-optimization. The feedback variants isolate whether the optional tracker is enabled.

4.2. Workload-level feedback design

The fast layer answers an immediate operational question: how should the terminal move the current-window instruction when the realized system is no longer following the planned path? Workload levels retain accumulated imbalance that a single-window flow can hide. The gate signal uses the waiting queue Q_t^g, while the yard signal uses total unfinished workload $N_t^y = Q_t^y + I_t^y$:

$$p_t^g = \mathrm{clip}\left(\frac{Q_t^g}{\kappa_g}, 0, \overline{p}_g\right), \tag{11}$$

and

$$p_t^y = \mathrm{clip}\left(\frac{Q_t^y + I_t^y}{\kappa_y}, 0, \overline{p}_y\right), \tag{12}$$

where I_t^y is the number of trucks currently in yard service. In the implemented one-truck-per-unit service model, it equals the number of busy yard units, so $Q_t^y + I_t^y$ has the physical meaning "trucks waiting or in service." The constants κ_g and κ_y scale the two signals, and clipping prevents an extreme observation from generating an unbounded action change. The diagnostic feedback vector is

$$F_t = \begin{pmatrix} p_t^g \\ p_t^y \end{pmatrix}. \tag{13}$$

Let $z^0 = (z^{0,q}, z^{0,y}) \in [0,1]^2$ be an externally specified normalized operating center whose coordinates index the appointment-quota and yard-capacity ranges. The quota coordinate is mapped over $[\underline{q}, \overline{q}]$; realized confirmations remain $a_t = \min\{R_t, q_t\}$. The center is a required policy input rather than an output of fast correction. An implementation may obtain it from a maintained predictive plan, a standard operating rule, or another documented baseline; its source and numerical value are part of the policy specification. The corresponding nominal physical instruction is

$$\Psi(z) = \begin{pmatrix} \underline{q} + z^q(\overline{q} - \underline{q}) \\ \underline{\mu}^y + z^y(\overline{\mu}^y - \underline{\mu}^y) \end{pmatrix}. \tag{14}$$

A workload correction of the form

$$b(F_t) = \begin{pmatrix} -k_{gg}p_t^g - k_{gy}p_t^y \\ k_{yg}p_t^g + k_{yy}(p_t^y - p_y^0) \end{pmatrix} \tag{15}$$

implements the desired operational signs. Gate and yard pressure both reduce the normalized appointment component. Yard pressure raises the normalized yard-capacity component, whereas low yard pressure allows capacity to return toward a less resource-intensive level. Gate pressure can also raise yard capacity because clearing the gate will soon generate downstream work.

The numerical implementation uses $\kappa_g = \kappa_y = 20$, $\overline{p}_g = \overline{p}_y = 1.5$, and

$$k_{gg} = 0.18, \quad k_{gy} = 0.28, \quad k_{yg} = 0.08, \quad k_{yy} = 0.45, \quad p_y^0 = 0.20. \tag{16}$$

These coefficients are design gains rather than estimates of a physical law. Their purpose is to encode a transparent feedback direction. The appointment response is stronger to yard pressure than to gate pressure because downstream congestion cannot be removed by gate processing alone, while yard capacity rises above a low workload reference. The present evidence does not identify an optimal gain vector. A separate queue-only reference diagnostic scales the gains between 0.5 and 1.5; it is a limited design check rather than validation of Eq. (12) or terminal-specific calibration.

Appointment-access protection is not introduced as a third control. The implemented action vector remains two-dimensional. Access is represented by the floor in Eq. (5), the unconfirmed-request share inside the congestion-and-access score V_t in Eq. (3), and separate reporting of confirmation, throughput, and unconfirmed requests. These devices protect the opportunity to obtain an appointment; they do not impose a completion or waiting-time guarantee.

The core state-responsive instruction is $\Psi(z^0 + b(F_t))$; Γ_t^{FB} then applies physical truncation, integer rounding, and the appointment-access floor. The normalized operating center z^0 is an externally supplied baseline input rather than an estimate inferred from F_t, while $b(F_t)$ supplies the execution-time correction. The optional tracker below replaces z^0 by an updated sequence z_n.

4.3. Optional sequential paired-perturbation tracker and safe execution

The optional paired-perturbation tracker asks which normalized operating center the fast correction should use. Starting from the documented input $z_0 = z^0$, it updates the center from two sequential one-window performance observations. This target is not the remaining-horizon objective in Eq. (4): it is the realized window cost observed after each perturbed action. Because the two actions are executed in consecutive windows rather than from cloned system states, their cost difference is a directional adaptation score, not an unbiased central finite-difference gradient of a fixed objective. Appointment consequences can also extend beyond the observation window through late arrivals and subsequent queues.

At iteration n, the policy selects coordinate $i_n \in \{1, 2\}$. Let e_{i_n} denote the associated unit vector, let $s \in \{-1, +1\}$, and let $\delta_n > 0$ be the perturbation radius. A normalized candidate is

$$\widetilde{z}_{n,s,t} = z_n + s\delta_n e_{i_n} + b(F_t). \tag{17}$$

The operational action is

$$\theta_{n,s,t} = \Gamma_t^{\mathrm{FB}} \left(\Psi\left(\Pi_{[0,1]^2}(\widetilde{z}_{n,s,t})\right)\right), \tag{18}$$

where $\Pi_{[0,1]^2}$ clips the normalized candidate and Γ_t^{FB} supplies the safe execution map. Let $\ell_{n,-}$ and $\ell_{n,+}$ be the realized weighted costs accumulated during the two consecutive decision windows after the negative and positive actions are implemented. The signed paired-perturbation score is

$$\widehat{d}_n = \frac{\ell_{n,+} - \ell_{n,-}}{2\delta_n} e_{i_n}. \tag{19}$$

The normalization by $2\delta_n$ records the intended separation before clipping and the integer execution mapping. It does not restore a gradient interpretation when the executable actions have unequal spacing or coincide. The score can also contain state-evolution, fixed-order, adjustment-cost, and fast-feedback effects because S_t^{fast} and F_t change between the two windows. These effects are treated as implementation bias rather than as zero-mean sampling noise.

The normalized center is updated according to

$$z_{n+1} = \Pi_{[0,1]^2}\left(z_n - \eta_n \widehat{d}_n\right), \tag{20}$$

where η_n is a diminishing learning rate. The implemented schedules are

$$\eta_n = \eta_0 (n+1)^{-0.6} \tag{21}$$

and

$$\delta_n = \max\{\delta_{\min}, \delta_0 (n+1)^{-1/3}\}. \tag{22}$$

The positive perturbation floor $\delta_{\min}$ prevents the nominal perturbation from vanishing. It may reduce, but cannot eliminate, coincident executed actions, especially for the discrete capacity coordinate. The archived vehicle-level implementation fully specifies the tracker execution sequence. At each policy reset, the center is initialized from the physical pair $(25, 8)$, and the first perturbed coordinate is drawn from the pseudo-random-number generator initialized by the recorded policy seed. Every pair is executed in negative–positive order, the center is updated after the positive-window cost is observed, and the next coordinate is then drawn from the same generator. The update operates from the beginning of the simulation; there is no separate warm-up update rule. Clipping, projection, and deterministic integer rounding are applied to every executed action, and pairs whose physical actions coincide after this mapping are retained rather than discarded. These specifications make the reported tracker instance reproducible, but they do not establish convergence of $\{z_n\}$, stabilization of the integer actions, or closed-loop queue stability.

When the optional tracker is enabled, its operational procedure is as follows.

1. For the archived vehicle-level experiments, initialize from $\theta^0 = (25, 8)$, so that $z_0 = z^0 = (13/18, 1/2)$.

2. Read the gate queue, yard queue, busy-yard count, and mean of up to six preceding request counts. The tracker additionally retains its center, coordinate, perturbation sign, and stored negative-perturbation cost as internal policy state.

3. Compute the fast workload correction in Eq. (15).

4. Perturb one coordinate of the normalized operating center, convert the candidate to a physical action, and enforce the safe feasible set.

5. Observe the realized one-window cost, apply the opposite perturbation in the next window, and observe the second cost.

6. Form the signed score in Eq. (19), update the center using Eq. (20), and select the next coordinate.

7. Repeat while reporting waiting, throughput, confirmation, resource use, and tail congestion separately.

The update is motivated by two-point stochastic approximation and online queue learning (Robbins and Monro, 1951; Kushner and Yin, 2003; Chen et al., 2026), but its sequential implementation does not satisfy the standard same-objective central-difference construction. Its operational role is narrower: it perturbs a low-dimensional baseline while the fast layer responds directly to workload and the execution map enforces feasible appointment and integer-capacity actions.

The methodological contribution consequently lies in the coupled gate–yard correction architecture rather than in a new finite-difference convergence result. The confirmation decision changes a future arrival pipeline, gate completions feed the downstream yard, and access outcomes must be reported with queue costs. The fast layer implements this operational coupling and constitutes the core method. The paired-perturbation tracker remains an optional extension, and its value is judged relative to the fast-only ablation rather than assumed from Eq. (19).

5. Conditional mechanism and model-fidelity properties

This section uses two accounting observations and five conditional results to clarify the mechanism; it does not present a solution to the complete terminal control problem. The sequence follows the reader's causal path: appointment uncertainty, gate-to-yard flow, local response information, PTO model fidelity, and safe execution.

5.1. Structural properties of the gate–yard map

Observation 1 (confirmation does not determine realized arrivals). Suppose that

$$\mathrm{Var}(A_t \mid a_t, H_t, C_t, E_t) = \sigma_t^2(a_t, H_t, C_t, E_t) > 0.$$

Then realized arrivals are not conditionally deterministic after the current decision and appointment-pipeline state are fixed. In particular, there is no deterministic function of a_t, H_t, C_t, and E_t that equals A_t almost surely.

Positive conditional variance rules out a deterministic mapping from the observed confirmation state to physical arrivals. A plan can regulate the arrival distribution, but it cannot eliminate no-shows, timing shifts, exception arrivals, or other compliance variation.

Observation 2 (gate-to-yard workload propagation). Equation (2) gives the exact accounting identity

$$N_{t+1}^y - N_t^y = D_t^g - D_t^y. \tag{23}$$

Hence $D_t^g > D_t^y$ implies $N_{t+1}^y > N_t^y$. Because actual completions satisfy $D_t^y \le Y_t(\mu_t^y)$, the stronger condition $D_t^g > Y_t(\mu_t^y)$ is sufficient, but not necessary, for yard workload to increase.

This accounting result does not imply that every gate-only intervention makes the yard worse than a fixed rule. It says only that realized gate output in excess of realized yard completions increases unfinished yard workload; potential capacity supplies an upper-bound comparison. Because I_t^y is bounded by the number of active units, persistent linear growth in $N_t^y = Q_t^y + I_t^y$ also implies persistent growth in the waiting queue. The present evidence cannot isolate the marginal value of joint gate–yard authority because it does not hold the feedback architecture fixed while changing only that authority.

5.2. Feedback identifiability and conditional queue-drift comparison

For feedback to assign corrective action to the appropriate operational lever, the next observed workload must respond distinguishably to appointment confirmation and yard capacity. Fix a predecision summary s and define the continuous-relaxation response map

$$\overline{F}^{+}(s,\theta) = \mathbb{E}[F_{t+1} \mid S_t^{\text{fast}} = s, \theta_t = \theta]$$

under a locally stationary operating regime. This one-step-ahead definition respects the decision clock: F_t is observed before θ_t is chosen. Let $F^\star$ be a desired workload target. A balanced point satisfies

$$\overline{F}^{+}(s^\star, \theta^\star) = F^\star. \tag{24}$$

Proposition 1 (local one-step response identifiability). Suppose that $\overline{F}^{+}(s^\star, \cdot) : \mathbb{R}^2 \to \mathbb{R}^2$ is continuously differentiable in a neighborhood of an interior continuous-relaxation point $\theta^\star$ satisfying Eq. (24), and that its action Jacobian

$$J_F(\theta^\star) = \left.\frac{\partial \overline{F}^{+}(s^\star, \theta)}{\partial \theta^\mathsf{T}}\right|_{\theta=\theta^\star}$$

is nonsingular. Then there are neighborhoods of $\theta^\star$ and $F^\star$ in which the relaxed action producing a given next-window mean feedback vector is locally unique and continuously dependent on that vector.

The proposition gives a precise local interpretation to "informative feedback." If the two actions have indistinguishable next-window effects, the Jacobian is singular or poorly conditioned. The result applies only before the clipping, rounding, and safe-execution operations in Eq. (18). It neither verifies the hand-set gains in Eq. (16) nor proves that the implemented rule reduces cost. Appendix Appendix A records the inverse-function argument.

We next state the direct queueing benefit that can be obtained when the feedback signs are operationally valid.

Proposition 2 (conditional reduction in yard-backlog drift). Fix the same full predecision state $\Xi_t = \xi$ for two counterfactual actions and evaluate both under the same conditional law of within-window exogenous events. Define the expected realized input–service imbalance

$$\Delta_t^\pi(\xi) = \mathbb{E}[D_t^{g,\pi} - D_t^{y,\pi} \mid \Xi_t = \xi, \theta_t = \theta_t^\pi], \qquad \pi \in \{\mathrm{PTO}, \mathrm{FB}\}.$$

Suppose $\Delta_t^{\mathrm{PTO}}(\xi) > 0$ and the feedback action satisfies

$$\Delta_t^{\mathrm{FB}}(\xi) \le \Delta_t^{\mathrm{PTO}}(\xi) - \gamma$$

for some $\gamma > 0$. Then

$$\mathbb{E}[N_{t+1}^{y,\mathrm{FB}} - N_t^y \mid \Xi_t = \xi] \le \mathbb{E}[N_{t+1}^{y,\mathrm{PTO}} - N_t^y \mid \Xi_t = \xi] - \gamma. \tag{25}$$

If $\Delta_t^{\mathrm{FB}}(\xi) \le 0$, the feedback action removes positive expected unfinished-yard-workload drift for that window.

Conditioning on Ξ_t fixes the residual-service state for the counterfactual comparison; it does not give the controller information beyond S_t^{fast}. The condition in Proposition 2 is substantive: the feedback action must reduce upstream pressure, increase realized downstream completions, or do both. Resource and access costs may offset the backlog improvement, so overall performance remains an empirical question. Appendix Appendix A verifies the one-step comparison.

The paired score does not define an estimable stochastic-approximation mean field in the present evidence. Conditional on the information before pair n, its deviation from any chosen reference field can always be partitioned into centered noise and conditional bias, but that accounting identity supplies no convergence argument. Here the bias includes state evolution, fixed perturbation order, changes in fast feedback, adjustment-cost asymmetry, finite perturbations, clipping, and the integer execution mapping. Standard projected stochastic approximation would require additional conditions that are neither imposed nor verified (Kushner and Yin, 2003; Borkar, 2008).

5.3. Model-fidelity and simulation-selection bounds for PTO

Let $J(u)$ be the expected cost of open-loop plan u under the true operating environment and $\widehat{J}(u)$ its expected cost under the PTO belief model. Define the uniform model-induced cost error over the candidate plan class as

$$\Delta_{\mathcal{U}} = \sup_{u \in \mathcal{U}} |J(u) - \widehat{J}(u)|. \tag{26}$$

Proposition 3 (exact-model PTO optimality within the open-loop class). If

$$J(u) = \widehat{J}(u), \qquad \forall u \in \mathcal{U},$$

and

$$u^{\text{PTO}} \in \arg\min_{u \in \mathcal{U}} \widehat{J}(u),$$

then

$$J(u^{\text{PTO}}) \leq J(u), \qquad \forall u \in \mathcal{U}. \tag{27}$$

This proposition identifies the population-objective benchmark. It is optimal within the specified open-loop plan class when the planning model reproduces true plan costs. Taking $\mathcal{U} = \mathcal{U}_{\text{pair}}$ gives the ideal fixed-pair specialization, but the implemented benchmark estimates plan costs by finite Monte Carlo simulation and therefore approximates this optimizer. Neither result establishes optimality over the full time-varying class in Eq. (10) or over closed-loop state-feedback policies outside $\mathcal{U}$. Appendix Appendix A records the direct argument.

Proposition 4 (PTO model-fidelity and finite-sample selection bounds). Let

$$u^{\mathrm{PTO}} \in \arg\min_{u\in\mathcal{U}} \widehat{J}(u), \qquad u^* \in \arg\min_{u\in\mathcal{U}} J(u),$$

and suppose $\Delta_{\mathcal{U}} < \infty$. Then

$$J(u^{\mathrm{PTO}}) - J(u^*) \le 2\Delta_{\mathcal{U}}. \tag{28}$$

For the finite-simulation selector, let $\widetilde{J}_N(u)$ be the Monte Carlo estimate of $\widehat{J}(u)$ based on planning sample size N, and let

$$u_N^{\mathrm{MC}} \in \arg\min_{u\in\mathcal{U}} \widetilde{J}_N(u).$$

On any event for which

$$\sup_{u\in\mathcal{U}} \left| \widetilde{J}_N(u) - \widehat{J}(u) \right| \le \varepsilon_N,$$

the selected plan satisfies

$$J(u_N^{\mathrm{MC}}) - J(u^*) \le 2\Delta_{\mathcal{U}} + 2\varepsilon_N. \tag{28a}$$

Equations (28)–(28a) separate model error from finite planning-sample error. In the implemented model-fidelity selector, $\mathcal{U}_{\mathrm{pair}}$ contains 95 fixed pairs: 19 integer quotas from 12 through 30 crossed with five integer yard-capacity levels from 6 through 10. Each candidate is evaluated on four common planning paths at each fidelity level. For zero-based scenario, fidelity-level, and planning-sample indices s, f, and n, the model seed is $70000 + 10000s + 1000f + n$, and the fixed-policy seed is $91000 + n$. Ties are resolved lexicographically by (estimated mean cost, quota, yard capacity). The archive retains all 380 candidate–fidelity records and the selected pairs. No numerical concentration radius ε_N is estimated, so Eq. (28a) remains a conditional deterministic bound rather than a finite-sample confidence guarantee for the reported selector. Appendix Appendix A gives both comparisons.

Proposition 5 (persistent realized completion imbalance under a maintained PTO instruction). Suppose a fixed-pair PTO instruction, possibly selected under an overestimated yard-capacity model, is maintained during the K-window comparison without online re-estimation or re-solving from realized completions. If the induced executed actions satisfy

$$\mathbb{E}[D_t^{g,\mathrm{PTO}} - D_t^{y,\mathrm{PTO}}] \ge \delta > 0, \tag{29}$$

then

$$\mathbb{E}[N^y_{t+1} - N^y_t] \geq \delta. \tag{30}$$

If the analogous condition holds in each of K consecutive windows under the maintained instruction, then

$$\mathbb{E}[N^y_{t+K} - N^y_t] \geq K\delta. \tag{31}$$

Proposition 5 does not derive Eq. (29) from capacity overestimation. It states the queue consequence once a maintained instruction induces realized gate output above realized yard completions. Because $D^{y,\mathrm{PTO}}_t \leq Y_t(\mu^{y,\mathrm{PTO}}_t)$, the stronger condition $\mathbb{E}[D^{g,\mathrm{PTO}}_t - Y_t(\mu^{y,\mathrm{PTO}}_t)] \geq \delta$ is sufficient, but not necessary, for Eq. (29). Capacity overestimation is one possible antecedent of this imbalance. Re-identification matters because it may change the executed action and terminate Eq. (29); the proposition does not claim that lack of re-identification is a logically necessary cause of drift. Appendix Appendix A gives the one-window and cumulative arguments.

5.4. Safe feasibility and limits of the analytical claim

The implemented execution map supplies the final operational guardrail. Let $\Theta^{\mathrm{FB},d}_t$ denote the implementable subset of Θ^{FB}_t after the prescribed integer appointment and discrete-capacity mappings. For any raw physical instruction v_t, the executed feedback action is

$$\theta_t = \Gamma^{\mathrm{FB}}_t(v_t) \in \Theta^{\mathrm{FB},d}_t, \tag{32}$$

provided that the joint feasible set is nonempty. The map clips the quota and capacity to their physical bounds, applies the pre-specified deterministic integer mapping, and enforces the compatible appointment-access floor in Eq. (5); current requests then determine $a_t = \min\{R_t, q_t\}$. Feasibility therefore follows from the construction of the executed map; Eq. (32) makes no nearest-Euclidean-point claim.

The execution map ensures action feasibility, not queue stability or economic optimality. Those outcomes additionally require sufficient capacity, a valid feedback direction, acceptable resource and adjustment costs, and the appointment-access safeguard. Appendix Appendix A records the membership argument. The analysis consequently supports the following conditional interpretation.

First, the exact-model PTO optimizer is optimal within its specified open-loop class; the model-fidelity Monte Carlo selector inherits that interpretation only up to the finite planning-sample error in Eq. (28a). The explore-then-select and rolling-surrogate implementations are empirical planning benchmarks rather than instances of that bound. Second, as model fidelity deteriorates, the open-loop guarantee weakens and a maintained instruction can produce positive downstream drift when realized gate output persistently exceeds realized yard completions. Third, a feedback action reduces that drift only when it actually lowers the realized input–service imbalance; resource use and appointment access must still be evaluated. Fourth, local response identifiability does not validate the selected gains, and the paired tracker has no established convergence guarantee.

These boundaries are central to the paper's contribution. The proposed method is valuable not because it defeats perfect predictive planning by assumption, but because port operations are exposed to compliance variation, service-time uncertainty, capacity disruptions, and evolving workload conditions. Feedback coordination supplies a transparent correction mechanism when the realized service ecosystem departs from the model on which the plan was based.

These results determine the order of the computational study. The experiments first compare representatives of the three policy families against the implemented fixed-pair PTO benchmark, then isolate model fidelity and use a separate cadence-triggered recovery diagnostic to compare post-warm-up persistence across trigger times, and finally examine appointment access and design sensitivity. This sequence tests when the restricted high-fidelity PTO benchmark remains strong and when workload-responsive correction can respond to a maintained realized imbalance.

6. Computational study: public-data-calibrated experiments and mechanism diagnostics

The computational study separates public-data-calibrated comparisons from independent mechanism diagnostics. The fixed rule represents non-responsive operation. High-fidelity and misspecified fixed-pair PTO represent predictive planning. A separate aggregate reconstruction evaluates delayed activation of a prespecified quota-recovery rule; it is not an adaptive or re-solved PTO policy. The feedback variants are fast-only correction and fast-plus-tracker feedback.

We label the calibration block E0 and the three policy experiments E1–E3. E0 separates calibrated inputs from simulation assumptions. E1 compares the three policy families under high-fidelity information and immediately asks whether the optional tracker improves fast correction. E2 then degrades only the PTO belief and follows the stress test with a separate aggregate reconstruction of recovery timing and feedback-component choice. E3 examines appointment access and the implemented congestion-and-access weighting, followed by an independent gain–lateness diagnostic. Each block has a distinct evidentiary role and data-generating pipeline.

6.1. Data sources and calibration logic

The calibrated E0–E3 branch uses public operational records to discipline the demand, compliance, service, and disruption mechanisms of the simulation; the gain–lateness analysis uses an independent aggregate reference implementation. The calibrated branch does not reconstruct a complete truck-level appointment–arrival–gate–yard trajectory. Instead, several complementary sources calibrate quantities observed at different levels of port operations. SC Ports publishes gate missions, transactions, turn times, crane productivity, vessel calls, pier moves, and related operating metrics (South Carolina Ports Authority, 2026). Port of Virginia weekly dashboards report reservation moves, missed reservations, reservation shares, and turn-time measures (Virginia Port Authority, 2026). Port Houston terminal-status reports provide completed transactions, truck turn times, yard utilization, and import-dwell indicators (Port Houston, 2026). Public Mendeley datasets supply truck-appointment instances, capacity-management instances, and container-terminal event logs (Huynh et al., 2018; Mar-Ortiz, 2020; Prasetyo et al., 2023). Table 2 states what each source contributes and, equally important, what it cannot identify.

E0 is the calibration step. It converts these observations into four workload scenarios. The median SC Ports day is used as the reference demand level; the low, high, and peak days scale that reference. The observed missed-reservation evidence is represented by a 4% missed-reservation rate, the Port Houston reports provide an approximate 38.9% yard-utilization reference, and the affected-window disruption probability is set to approximately 6.1%. Table 3 summarizes the resulting scenarios. The 0.989–1.069 workload multipliers describe a narrow calibrated operating range rather than a broad demand stress envelope; E2 supplies the separate model-fidelity stress test. Tables 2–4 distinguish public-data anchors and calibrated proxies from simulation design assumptions.

No single public dataset contains every layer of the modeled process, so the calibration combines sources. SC Ports anchors workload variation; Port of Virginia anchors reservation compliance; Port Houston anchors yard pressure and disruption; and the benchmark and event-log data support capacity ranges and the gate–yard process sequence. The combined evidence improves operational plausibility but does not remove the limitations of aggregate

Table 2: Calibration audit: evidence used to discipline the simulation.

Evidence source	Observable scale	Role in the simulation	Limitation
SC Ports operational records	Daily gate missions and terminal performance indicators	Anchor the reference workload and the low-to-peak demand multipliers	Aggregate records do not link individual appointments, arrivals, and yard services
Port of Virginia weekly reports	Reservation moves, missed reservations, reservation shares, and turn times	Discipline appointment appearance and non-compliance assumptions	Weekly aggregation masks intraday compliance patterns and terminal heterogeneity
Port Houston terminal reports	Turn times, yard utilization, dwell, and reported disruptions	Anchor yard-pressure and capacity-disruption mechanisms	Yard utilization is a system-level proxy rather than vehicle-level service capacity
Public benchmark and event-log datasets	Appointment-capacity instances and truck-in, truck-out, stack, and unstack events	Support feasible ranges and the gate-to-yard process sequence	Research datasets do not identify counterfactual outcomes under the tested policies
Simulation design assumptions	Decision-window length, confirmation cutoff, intraday lateness split, service-time distributions, server counts, and action bounds	Complete the vehicle-level stochastic environment where public records are silent	These values support mechanism analysis and are not terminal-specific estimates

Table 3: Public-data-calibrated workload scenarios used in calibration and the high-fidelity policy comparison.

Scenario	Missions	Multiplier	Requests/window	Disruption
Low	5,961	0.989	22.75	0.061
Median	6,026	1.000	23.00	0.061
High	6,401	1.062	24.43	0.061
Peak	6,444	1.069	24.60	0.061

data. The appropriate interpretation is external calibration and mechanism-oriented stress testing, not causal evaluation of a policy deployed at a specific terminal.

6.2. Simulation design, policy variants, and statistical comparison

The vehicle-level discrete-event simulation uses 30-minute decision windows over a 30-day horizon, with the first seven days treated as warm-up. Its appointment-outcome probabilities are 0.7872 on time, 0.1152 one window late, 0.0576 two windows late, and 0.04 no-show. The aggregate reservation reports discipline the 0.04 missed-reservation share. Conditional on appearance, the remaining 0.96 mass is split in the design proportions 82:12:6; the public records do not identify that intraday split or the two-window support. Exception arrivals enter independently at 0.5 trucks per window. A probability-0.10 arrival shock multiplies the window request rate by 1.25, with normalization that preserves its configured mean. Gate and yard service times are lognormal. With probability 0.061, a disruption removes one instructed yard unit for the affected window, subject to the six-unit floor. Tables 4 and 5 consolidate the parameter values, units, and evidentiary status so that calibrated inputs are not confused with design assumptions.

Table 6 separates the exactly reproducible vehicle-level blocks from the independently reconstructed diagnostics and records the remaining evidence limits.

Within the archived vehicle-level comparisons, competing policies receive common random numbers for appointment requests, appearance outcomes, service times, exception arrivals, and disruptions. E1–E3 use fast-plus-tracker feedback, so their state-responsive results belong to that combined implementation. The original vehicle-level E2 pipeline does not evaluate fast-only correction, and its misspecification result is therefore not attributed to the fast component alone. A high-fidelity component diagnostic and the independent aggregate severe-error reconstruction compare fast-only with fast-plus-tracker feedback under their respective implementations; the reconstruction shares starting seeds but does not use event-indexed common-random-number paths. Performance is assessed using weighted cost, waiting, yard congestion, throughput, confirmation, unconfirmed requests, resource use, terminal backlog, and late-horizon queue drift.

For E1, E3, and the high-fidelity component diagnostic, zero-based scenario, replication, and policy indices s, r, and p use scenario seed $10000(s+1)+r$ and policy seed $10000(s+1)+r+100000(p+1)$. E2 evaluation uses scenario seed $30000(s+1)+r$, feedback-policy seed equal to that scenario seed plus 200000, and PTO-policy seed equal to the scenario seed plus $100000+10000f$, where f indexes fidelity. Competing policies evaluated in a

Table 4: Core simulation parameters. "Calibrated proxy" denotes a value disciplined by aggregate public evidence but not identified from a complete terminal-level trajectory.

Component	Parameter	Value	Unit/status	Role or evidence
Time scale	Window; horizon; warm-up	30; 30; 7	minutes; days; days / assumed	Decision frequency, horizon, and seven-day metric exclusion
Gate service	Mean; coefficient of variation; lanes	5; 0.35; 4	minutes; ratio; lanes / assumed	Defines nominal gate processing capacity
Yard service	Mean; coefficient of variation; active units	10; 0.50; 6–10	minutes; ratio; units / assumed	Creates the downstream capacity range
Appointments	Quota range; missed-reservation probability	12–30; 0.04	trucks/window; probability	Action bounds; calibrated proxy from reservation reports
Appointment timing	On time; late 1; late 2; no-show	0.7872; 0.1152; 0.0576; 0.04	probabilities / mixed	Missed share is calibrated; 82:12:6 conditional appearance split is assumed
Lateness support	Base $L = 2$; maximum lateness	2; 60	windows; minutes / assumed	Reference diagnostic also tests $L = 1$ and $L = 4$
Unbooked demand	Exception-arrival mean	0.5	trucks/window / assumed	Independent arrivals outside appointment requests
Arrival shock	Probability; multiplier	0.10; 1.25	probability; factor / assumed	Rate normalized to preserve the configured request mean
Disruption	Probability; capacity effect	0.061; minus one	probability; yard unit / mixed	Temporary loss subject to the six-unit floor

Table 5: Policy and statistical parameters.

Component	Parameter	Value	Role
Vehicle-level base cost	W_0	0.30, 0.30, 0.20, 0.10, 0.10	Gate queue, yard queue, effective yard units, quota change, and congestion-and-access score
Cost sensitivity	W_1; W_2	0.25, 0.25, 0.15, 0.10, 0.25; 0.20, 0.20, 0.10, 0.10, 0.40	Raises the combined congestion-and-access weight; this is not an access-only reweighting
Appointment access	$\rho_{\min}$; W_R; $\overline{R}_1$	0.85; 6; 25	Feedback-specific rolling quota floor, lookback, and empty-history initialization
Fast-feedback scaling	κ_g, κ_y; $\overline{p}_g, \overline{p}_y$; p_y^0	20, 20; 1.5, 1.5; 0.20	Queue scaling, clipping, and yard reference
Fast-feedback gains	$k_{gg}, k_{gy}, k_{yg}, k_{yy}$	0.18, 0.28, 0.08, 0.45	Design gains used in Eq. (16)
Paired update	$\eta_0, \delta_0, \delta_{\min}$	0.08, 0.10, 0.03	Step size and perturbation sequence
Explore-then-select PTO	Exploration share; candidate grid	0.20; $19 \times 5 = 95$	Six-day exploration within warm-up, then surrogate pair selection
Vehicle-level baseline	θ^0; z^0	$(25, 8)$; $(13/18, 1/2)$	Fixed-rule pair and initial center for vehicle-level fast-only and fast-plus-tracker policies
Aggregate-diagnostic center	z^0; θ^0	$(0.50, 0.60)$; nominal $(21, 8)$	Separate ex ante center used only in the independent aggregate reconstructions
Replication	E1; E2; component diagnostic; E3	50; 50; 50; 12	Common-random-number paired comparisons in the vehicle-level blocks
Inference	E2 paired cost comparison	2,000 bootstrap resamples	Paired 95% confidence intervals; aggregate seed-block intervals use the Student-t procedure stated in Table 10

Note: Weight order is w_g, w_y, w_r, w_s, w_v. The resource, switching, and congestion-and-access terms are $\widetilde{m}_t/10$, $|q_t - q_{t-1}|/18$, and V_t; yard-capacity changes are not separately penalized. Because V_t combines a queue threshold with five times the unconfirmed share, W_1 and W_2 are joint congestion-and-access sensitivities. The aggregate center maps to $(21, 8.4)$ before rounding and $(21, 8)$ afterward and is not used in the vehicle-level experiments.

Table 6: Computational reproducibility coverage and evidence scope.

Block	Public archive coverage	Remaining evidence limit
E0 calibration	Paper-date processed inputs, source and role manifests, download and replay scripts, hashes, and a retained 2026-07-09 raw snapshot	The third-party raw snapshot is not publicly redistributed; live dashboards roll forward and need not reproduce the paper-date bytes
Vehicle-level simulation	Python source for first-come-first-served event logic, residual service across windows, minute-level queue-area accounting, and the two-day drain	Calibration records are aggregate and do not constitute vehicle-level field validation
Cost and access	Implemented normalizations and weights, fixed pair $(25, 8)$, six-window request lookback, and empty-history initialization at 25	The normalized performance index is neither money nor a social-welfare measure
Feedback execution	Initial center, gains, seeded coordinate draws, negative–positive order, update timing, projection, and integer rounding	Alternative-initializer sensitivity and tracker convergence are not established
PTO selection	95 fixed-pair candidates, four planning paths per candidate, deterministic seeds, lexicographic tie-break, and all 380 E2 search records	Finite planning-sample selection error is not numerically bounded
E1–E3 and component diagnostic	Generation scripts, raw outputs, and deterministic verification checks for every reported vehicle-level block	E2 contains no matched fast-only arm
Tables 10 and 12	Reconstruction code, configurations, seeds, path-level outputs, and seed-block summaries	Independent aggregate diagnostics; they are not vehicle-level E2 outputs and are not pooled with them

scenario–replication block share the generated request, appearance, service-time, exception-arrival, and disruption paths; policy-specific randomization remains deterministic.

The archived vehicle-level fixed rule uses the physical pair $(25, 8)$. Both vehicle-level feedback specifications initialize from that same pair, which corresponds under Eq. (14) to $z^0 = (13/18, 1/2)$. This initializer is held fixed across E1–E3 and the high-fidelity component diagnostic. The independent aggregate diagnostics use the separate ex ante center $z^0 = (0.50, 0.60)$, which maps to $(21, 8.4)$ before integer rounding and $(21, 8)$ afterward. Those diagnostic centers are not used to infer the vehicle-level initializer. Sensitivity to alternative vehicle-level starting centers has not been tested.

The reported confirmation rate is post-warm-up confirmations divided by post-warm-up appointment requests. Throughput is the number of trucks arriving during the post-warm-up arrival window and completing by the drain limit, divided by post-warm-up appointment requests. Its numerator can include exception trucks and appointment-pipeline spillovers, so throughput is not a cohort-specific completion probability. Within each replication, P95 waiting is computed over eligible trucks that complete both services; reported tables average the replication-level P95 values.

6.3. High-fidelity policy comparison and component choice

E1 asks whether the evaluated fast-plus-tracker implementation improves on a non-responsive rule and how it compares with a high-fidelity fixed-pair planner. Averaged across the four calibrated workloads, its weighted cost is 81.8% below the fixed rule, and mean total waiting falls from 175.6 to 15.8 minutes.

Table 7: E1: policy performance averaged across four public-data-calibrated workload scenarios.

Policy	Cost	Mean wait (min)	P95 wait (min)	Throughput	Confirmation
Fixed rule	2.168	175.6	278.1	0.912	0.929
Fast-plus-tracker feedback	0.394	15.8	39.9	0.883	0.898
Explore-then-select fixed-pair PTO	0.296	4.2	11.9	0.817	0.829

Note: PTO denotes predict-then-optimize; P95 denotes the 95th percentile. Within each replication, P95 is the empirical quantile of eligible completed vehicles; the table then averages those replication-level quantiles. The archived 800-row E1 output also contains a gate-only heuristic. It is omitted here because it uses a different initial center, a gate-only observed cost, and a fixed eight-unit yard instruction, so it is not a matched joint-authority ablation.

Table 7 shows that the explore-then-select fixed-pair PTO implementation has the lowest reported weighted cost and waiting among the tested representatives. It is an empirical planning benchmark: unlike the E2 selector, it is not a Monte Carlo minimizer covered by Eq. (28a). Part of its advantage comes from confirming fewer requests: fast-plus-tracker feedback has a 6.9 percentage-point higher confirmation rate and a 6.6 percentage-point higher throughput rate. E1 therefore establishes a comparison between the evaluated policy representatives, not a dominance ranking over policy families.

The component diagnostic next asks whether the optional tracker improves fast correction under the same high-fidelity information structure. Fixed-pair PTO provides a planning reference; fast-only and fast-plus-tracker feedback share the direct correction rule and differ only in whether the tracker is enabled.

Table 8: High-fidelity planning and feedback-component diagnostic, 50 replications. This is not a misspecification test.

Scenario	Policy	Cost	P95 wait (min)	Tail unfinished-system drift	Unstable share
High	Explore-then-select PTO	0.306	13.9	−0.006182	0.00
High	24-hour rolling-surrogate PTO	0.310	14.9	−0.008091	0.02
High	3-hour rolling-surrogate PTO	0.315	15.3	−0.009818	0.02
High	Fast-only feedback	0.381	41.5	−0.033182	0.34
High	Fast-plus-tracker feedback	0.463	52.1	−0.043727	0.40
Peak	Explore-then-select PTO	0.325	18.6	−0.010636	0.00
Peak	24-hour rolling-surrogate PTO	0.322	18.3	−0.007182	0.00
Peak	3-hour rolling-surrogate PTO	0.327	19.3	−0.011364	0.00
Peak	Fast-only feedback	0.391	42.7	−0.034818	0.24
Peak	Fast-plus-tracker feedback	0.492	53.9	−0.050727	0.30

Note: PTO denotes predict-then-optimize; P95 is computed within each replication and then averaged. The rolling-surrogate update windows are defined in Section 4.1; neither variant updates service capacity. Tail unfinished-system drift is the mean one-window change in waiting plus in-service trucks over the final fifth of evaluated windows. "Unstable" is the separate classification
$1\{\text{fitted tail slope} > 0.02 \text{ or terminal system} > 40\}$, where the fitted slope uses the average gate and yard waiting queues.

Table 8 reports the lowest cost, P95 waiting, and unstable share for the planning variants. Enabling the paired tracker makes late-horizon net drift more negative than fast-only correction, but it also raises cost, P95 waiting, and the unstable-classification share. The point estimates therefore provide no evidence of incremental net benefit from the tracker, and fast-only is the more parsimonious component choice under the tested high-fidelity conditions. This descriptive ordering is not extrapolated to the original vehicle-level misspecification pipeline; the independent aggregate severe-error reconstruction below provides a second, separately implemented component comparison.

6.4. Model misspecification and recovery timing

E2 asks what happens when a predictive planner maintains an outdated service model near the capacity boundary. The public-data-calibrated peak environment remains fixed; only the belief used by fixed-pair PTO changes. High fidelity uses unbiased request-volume and service-time factors. Mild misspecification multiplies assumed gate and yard service times by 0.90 and 0.88 and the effective request/appearance volume used in planning by 0.95; the corresponding factors are 0.75, 0.70, and 0.92 under medium misspecification and 0.58, 0.52, and 0.90 under severe misspecification. The appearance-probability vector itself is unchanged. The archived Monte Carlo selector chooses $(20, 8)$, $(23, 8)$, $(28, 7)$, and $(30, 6)$ under high fidelity, mild, medium, and severe misspecification, respectively, and then evaluates each pair in the unchanged true environment without re-identification. The fast-plus-tracker reference is held fixed and never receives the distorted PTO belief. It starts every replication from the archived physical pair $(25, 8)$, equivalently $z^0 = (13/18, 1/2)$; the same initializer and feedback settings are used at all four PTO fidelity levels. The experiment does not test sensitivity to alternative feedback initializers.

E2 is a fixed-pair no-reidentification stress test for model maintenance, not a rolling-horizon or certainty-equivalent PTO using the same observations to update and re-solve. The subsequent aggregate diagnostic is motivated by the same downstream-overload mechanism but asks a narrower question about trigger timing, post-warm-up persistence, and component choice. It does not estimate model parameters or re-solve a planning problem.

Table 9: E2: fixed-pair no-reidentification predict-then-optimize (PTO) model-fidelity stress test in the calibrated peak-workload environment. The feedback reference is fast-plus-tracker feedback. The paired cost difference is PTO minus feedback; a positive value favors feedback.

PTO belief	PTO cost	Fast-plus-tracker cost	PTO P95 (min)	Feedback P95 (min)	PTO–feedback cost and 95% CI
High fidelity	0.288	0.455	11.0	49.8	-0.167 $[-0.191, -0.145]$
Mild misspecification	0.380	0.455	36.3	49.8	-0.075 $[-0.097, -0.054]$
Medium misspecification	43.858	0.455	5,967	49.8	43.403 $[42.937, 43.848]$
Severe misspecification	85.372	0.455	11,601	49.8	84.918 $[84.430, 85.385]$

Note: P95 denotes the 95th percentile and CI denotes confidence interval. Each P95 entry averages 50 replication-level empirical P95 values.

Table 9 identifies a clear fidelity-dependent ranking within this no-reidentification design, and Figure 2 displays its cost and waiting escalation. Under high fidelity, PTO has a lower cost estimate and its paired 95% interval for PTO minus feedback lies below zero; the same pattern remains under mild misspecification, so a small model error does not reverse the ranking. The change occurs under medium misspecification: PTO cost rises to 43.858 and P95 waiting to 5,967 minutes, whereas feedback cost remains 0.455 and P95 waiting remains 49.8 minutes. Under severe misspecification, PTO cost reaches 85.372 and P95 waiting exceeds 11,600 minutes. The selected no-reidentification action admits trucks faster than the true yard can clear them; the resulting values do not describe adaptive PTO after capacity is re-estimated.

The throughput results clarify this mechanism. Under medium misspecification, PTO confirms 95.7% of requests and completes a throughput share of 86.8%, yet a large unresolved queue accumulates. Under severe misspecification, PTO throughput falls to 71.9%, compared with 86.3% for the fast-plus-tracker reference. High confirmation is therefore not evidence of service clearance when planned capacity is overstated.

The extreme P95 values are auditable stress statistics, not ordinary port turn-time estimates. After the 30-day arrival horizon, the simulator admits no new trucks and drains the system for at most two additional days, with all 10 yard units active and no new disruption draws. P95 is computed from vehicles that arrive after the seven-day warm-up and before the end of the arrival horizon and subsequently complete both services. Vehicles still unfinished at the drain limit are excluded from the waiting quantile and reported separately. Under medium and severe misspecification, every replication reaches the two-day drain limit with average unfinished counts of 1,951.58 and 6,448.42 trucks, respectively. The 11,601-minute value therefore indicates an unstable stress regime rather than a normal service level, and exclusion of unfinished vehicles can understate the most severe tail. The documented independent reconstruction reports terminal backlog and late-horizon drift directly, but its absolute values are not pooled with E2.

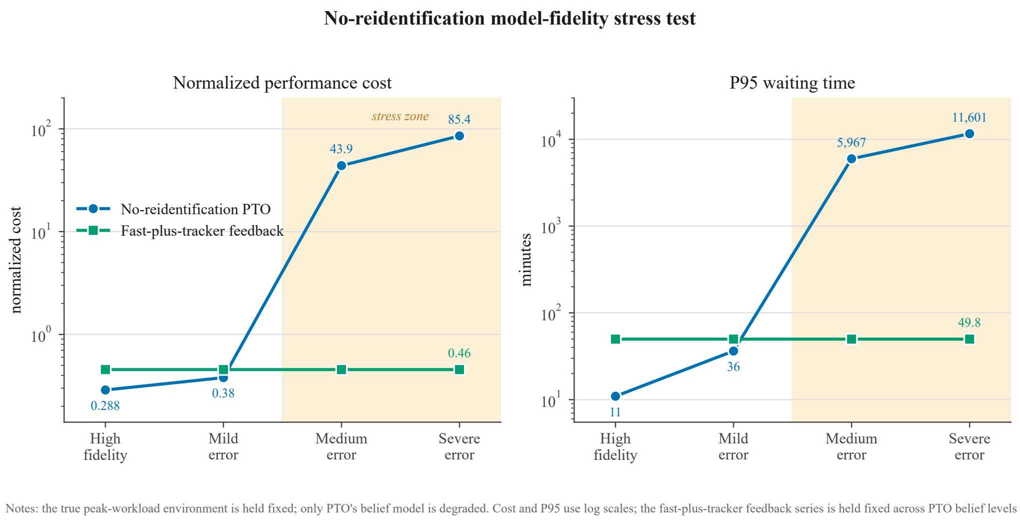

Figure 2: Fixed-pair no-reidentification model-fidelity stress test. The true peak-workload environment is fixed while only the belief used to select the maintained predict-then-optimize (PTO) pair is degraded. The feedback series is the evaluated fast-plus-tracker reference; a separately implemented cadence-triggered recovery diagnostic is examined below. P95 denotes the 95th percentile.

To examine response delay and component choice without presenting the result as a rerun of E2, we use a documented independent aggregate reconstruction motivated by the same downstream-overload mechanism. It uses end-of-window aggregate queues, the appearance probabilities $(0.80, 0.13, 0.03, 0.04)$, and gamma–Poisson gate and yard capacities with variance-to-mean ratio 1.8; these approximations differ from the vehicle-level event process. Saturated demand fills every offered appointment quota. In the action-revision branch, the reconstruction maintains the imposed pair $(24, 6)$, where the first entry is the appointment quota and the second is the number of instructed yard units, until a trigger after one, three, or seven days. Once triggered, it applies $(14, 6)$ while the current yard queue exceeds five trucks and $(15, 6)$ otherwise. A separate fixed-pair reference maintains $(15, 6)$ throughout. Although the configuration records the severe E2 service-time factors 0.58 and 0.52 as scenario provenance, those factors do not enter an estimator or optimization routine. The diagnostic neither estimates service time from realized completions nor searches or re-solves over $\mathcal{U}_{\text{pair}}$. Consequently, all displayed pairs and the five-truck threshold are reconstruction design settings, not actions selected by the vehicle-level E2 planner. The two aggregate feedback variants use the same operating center and gains, but their yard pressure uses the waiting queue alone and their appointment-access floor uses the scenario request mean rather than the busy-yard term in Eq. (12) and the rolling mean in Eq. (5). They are therefore component diagnostics, not literal reimplementations of the vehicle-level fast controller. All comparisons remain within this separate pipeline and are not pooled with Table 9.

The reconstruction has a separate aggregate diagnostic cost. With $\bar{R} = 24.595$, instructed yard units m_t, and end-of-window queues $Q^g_{t,\mathrm{end}}, Q^y_{t,\mathrm{end}}$, it is

$$
\begin{aligned}
c^{\mathrm{agg}}_t = {} & 0.30\frac{Q^g_{t,\mathrm{end}}}{20} + 0.30\frac{Q^y_{t,\mathrm{end}}}{20} + 0.20\frac{m_t - 6}{4} \\
& + 0.10\left[\frac{1}{2}\left(\frac{|q_t - q_{t-1}|}{18} + \frac{|m_t - m_{t-1}|}{4}\right)\right] + 0.10\frac{[\bar{R} - a_t]_+}{\bar{R}}.
\end{aligned}
$$

Unlike Eq. (3), this diagnostic cost uses end-of-window rather than minute-average queues, includes changes in both quota and instructed yard units, and contains no congestion-threshold indicator. It is reported only within Tables 10 and 12.

Table 10 and Figure 3 report this reconstruction. The trigger clock starts at simulation day zero, while diagnostic cost is averaged only after the seven-day warm-up. Thus, the one- and three-day branches activate during warm-up and have six and four days, respectively, to dissipate backlog before evaluation; the seven-day branch activates at the evaluation boundary. The entries measure post-warm-up persistence under those trajectories, not cumulative cost from day zero or a clean common-state comparison of activation delays.

Maintaining $(24, 6)$ yields aggregate diagnostic cost 73.884, terminal yard backlog of approximately 7,931 trucks, and positive late-horizon drift of 5.526 trucks per window. The one- and three-day branches have post-warm-up costs of 0.078 and 0.094 and terminal backlogs of approximately 2.2–2.3 trucks. The seven-day branch eventually removes persistent positive drift and ends with a backlog near 2.0 trucks, but its evaluation-period cost remains 5.921. The always-on $(15, 6)$ reference has cost 0.084 and terminal backlog 3.0 trucks. Fast-only feedback has cost 0.226, backlog 2.4 trucks, and drift -0.000081, whereas fast-plus-tracker feedback has cost 0.237, backlog 4.4 trucks, and drift 0.000537. For fast-only minus fast-plus-tracker, the seed-block mean differences are -0.0111 in cost, -1.918 trucks in terminal backlog, and -0.000618 trucks per window in drift; the three pointwise 95% intervals exclude zero. All policies other than the maintained overloaded pair have late-horizon drift below 0.001 trucks per window in absolute value. These comparisons belong only to the aggregate reconstruction and are not a continuation of, or matched fast-only ablation within, vehicle-level E2.

The seven-day branch ends with a small terminal backlog but retains a large post-warm-up cost because it enters evaluation with an accumulated queue. The ordering shows that earlier activation leaves less congestion at the common evaluation boundary. It does not estimate the from-start cost of alternative response delays, quantify statistical capacity re-identification, implement rolling-horizon updating, or establish an advantage of feedback over a promptly re-solved planner.

Table 10: Documented independent aggregate severe-overload reconstruction: post-warm-up outcomes under an imposed action pair, delayed activation of a prespecified quota-recovery rule, and two feedback specifications.

Setting	Policy or rule	Aggregate diagnostic cost	Terminal yard backlog	Late-horizon drift
Action reference	Maintained pair $(24, 6)$	73.884	7,931.2	5.526
Recovery rule	Trigger after 7 days	5.921	2.0	−0.000111
Recovery rule	Trigger after 3 days	0.094	2.3	0.000246
Recovery rule	Trigger after 1 day	0.078	2.2	0.000254
Feedback reference	Fast-only feedback	0.226	2.4	−0.000081
Feedback reference	Fast-plus-tracker feedback	0.237	4.4	0.000537
Fixed-pair reference	Maintained pair $(15, 6)$	0.084	3.0	−0.000004

Fast-only minus fast-plus-tracker seed-block comparison

	Aggregate diagnostic cost	Terminal yard backlog	Late-horizon drift
Mean difference	−0.0111	−1.918	−0.000618
Pointwise 95% interval	[−0.0129, −0.0093]	[−2.373, −1.462]	[−0.000916, −0.000319]

Note: Action pairs list appointment quota and instructed yard units. Backlog is in trucks; drift is the ordinary-least-squares slope of mean yard queue over the final 10 days, in trucks per window. The comparison is fast-only minus fast-plus-tracker. Brackets are pointwise two-sided 95% Student-t intervals across 10 starting-seed-block differences (9 degrees of freedom), with 40 vectorized paths per block and no multiplicity adjustment. Common starting seeds are not event-indexed common-random-number paths; the intervals cover simulation-seed variability within this reconstruction, not model, calibration, parameter, or field uncertainty.

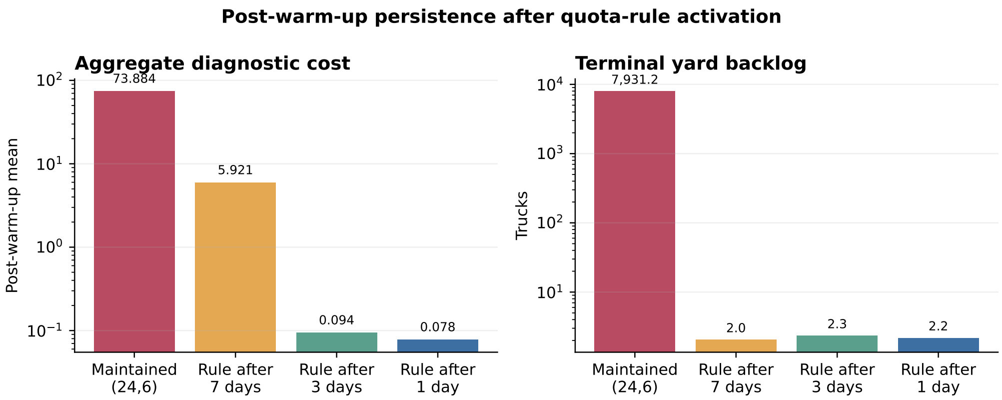

Figure 3: Recovery from an imposed overloaded pair in the documented independent aggregate reconstruction. Bars show post-warm-up mean aggregate diagnostic cost (left) and terminal yard backlog (right). Triggers are counted from simulation day zero, so the one- and three-day rules activate during the seven-day warm-up and the seven-day rule activates at the evaluation boundary. Both vertical axes use logarithmic scales. Feedback references appear in Table 10 but are not plotted.

6.5. Appointment-access trade-offs and robustness checks

E3 asks whether lower waiting is accompanied by more restrictive appointment access and how the reported ordering changes when the congestion-and-access weight is raised. The reruns change vehicle-level cost accounting and the composite cost observed by the paired tracker. The explore-then-select PTO surrogate, however, retains its hard-coded scoring coefficients and does not read W_1 or W_2. E3 is therefore an accounting-and-tracker-response sensitivity, not a common-objective reoptimization of every policy. Under W_0, fast-plus-tracker feedback confirms 88.6% of high-workload requests and 88.1% of peak-workload requests, compared with 82.7% and 82.8% for PTO, leaving approximately 1,594 and 1,453 fewer requests unconfirmed. The archived $(\text{PTO}, \text{feedback})$ cost pairs under W_1 are $(0.363, 0.608)$ for high workload and $(0.368, 0.612)$ for peak workload; under W_2, they are $(0.416, 0.620)$ and $(0.418, 0.688)$. This is a cost–access trade-off within the implemented specifications, not an access-only effect: increasing w_v raises both the congestion-threshold indicator and the unconfirmed-request share inside V_t.

Table 11 reports the base-weight outcomes used for a complementary post hoc access valuation. Let $C_\pi^\lambda = C_\pi + \lambda(1 - \alpha_\pi)$, where C_π is the weighted cost under W_0 and α_π is the confirmation rate. Because C_π already contains the base congestion-and-access term $w_v V_t$, λ is an additional horizon-level charge on unconfirmed demand; it is not the weight w_v used in the W_1 and W_2 reruns. When $\alpha_{\text{FB}} > \alpha_{\text{PTO}}$, the post hoc indifference threshold is

Table 11: E3: appointment-access trade-off under the vehicle-level base weight vector W_0.

Scenario	Policy	Cost	Confirmation	Throughput	Unconfirmed
High	Fast-plus-tracker feedback	0.468	0.886	0.871	3,082
High	Explore-then-select PTO	0.310	0.827	0.814	4,676
Peak	Fast-plus-tracker feedback	0.496	0.881	0.867	3,222
Peak	Explore-then-select PTO	0.318	0.828	0.815	4,675

Note: PTO denotes predict-then-optimize. The base vector is $W_0 = (0.30, 0.30, 0.20, 0.10, 0.10)$, ordered as $(w_g, w_y, w_r, w_s, w_v)$.

$$\lambda^* = \frac{C_{\text{FB}} - C_{\text{PTO}}}{\alpha_{\text{FB}} - \alpha_{\text{PTO}}}.$$

Using the archived outputs, $\lambda^* = 2.67$ in the high workload and 3.32 in the peak workload. This calculation holds the observed policy outcomes fixed and is expressed in normalized performance-cost units, not money. It therefore illustrates the additional access value required to reverse the reported ordering but does not estimate a social or contractual valuation.

The final diagnostic tests two unestimated design choices in a documented independent aggregate reconstruction with five seed blocks and 80 simulated paths per block. It scales all four gains in Eq. (16) from 0.5 to 1.5 and redistributes the 16% late-arrival mass over supports of one, two, and four windows. Relative to Eq. (12), this reference code uses queue-only yard pressure by setting the busy-yard term to zero; relative to Eq. (5), it replaces the rolling request average by the scenario mean. It is therefore a sensitivity check for a queue-only variant of the fast rule, not a replay of the vehicle-level controller.

Table 12: Documented independent aggregate reconstruction of gain and appointment-lateness sensitivity for the queue-only fast-feedback variant. Cost is the aggregate diagnostic index defined above; queue and terminal-backlog ranges are in trucks, and drift is in trucks per window.

Scenario	Design change	Tested values	Diagnostic cost range	Mean queue range	Terminal yard range	Max. $\lvert\text{drift}\rvert$
High	Gain multiplier	0.50–1.50	0.2240–0.2262	6.08–6.50	2.67–3.33	0.000135
Peak	Gain multiplier	0.50–1.50	0.2246–0.2267	6.08–6.50	2.67–3.33	0.000135
High	Lateness support L	1, 2, 4	0.2240–0.2250	6.22–6.28	2.59–3.11	0.000156
Peak	Lateness support L	1, 2, 4	0.2246–0.2256	6.22–6.28	2.59–3.11	0.000156

Note: L is the maximum modeled lateness in decision windows. The reconstruction uses the same horizon, warm-up, service scales, resource bounds, target access fraction, and high/peak request rates, but approximates yard pressure with the waiting queue alone and the rolling access floor with the scenario request mean. Each range is computed from path-level output under fixed seeds. Drift is the ordinary-least-squares slope of the mean yard queue over the final 10 days.

Across the tested gain multipliers, aggregate diagnostic cost varies by less than 1% of the base value and mean queues range from 6.08 to 6.50 trucks. Changing lateness support from one to four windows produces cost ranges of 0.2240–0.2250 in the high scenario and 0.2246–0.2256 in the peak scenario, with mean queues between 6.22 and 6.28 trucks. The maximum absolute drift is 0.000156 trucks per window. These results show no knife-edge reversal for the queue-only reference variant over the tested range. They do not validate Eq. (12)'s busy-yard signal, identify an optimal gain vector or true lateness distribution, or by themselves test fast-only correction in the vehicle-level E2 pipeline.

7. Discussion and managerial implications

7.1. When predictive planning remains reliable

The evidence favors predictive planning as the operating backbone when the maintained model ranks feasible plans accurately. In E1, the explore-then-select fixed-pair implementation has the lowest weighted cost and waiting among the reported representatives. In E2, the separate Monte Carlo selector also favors high-fidelity planning; only that selector is subject to the finite planning-sample term in Eq. (28a). The aggregate reconstruction adds a narrower persistence diagnostic: rules triggered after one or three days are already active before the seven-day evaluation boundary and enter it with little residual backlog. Because this is not a common-state delay comparison and the diagnostic neither estimates a model parameter nor solves an updated plan, it does not report the performance of re-identified or rolling-horizon PTO. The managerial question remains whether model maintenance and feasible action revision occur before congestion builds.

The component evidence also discourages unnecessary algorithmic complexity. In the vehicle-level diagnostic, enabling the paired tracker makes late-horizon drift more negative but produces higher cost, P95 waiting, and unstable-classification shares than fast-only correction. In the separate aggregate reconstruction, fast-only has lower cost, terminal backlog, and drift, and the pointwise seed-block intervals exclude zero. Those intervals quantify simulation-seed variability only and do not turn the reconstruction into an event-matched or vehicle-level comparison. The tracker remains an exploratory option; the simpler fast rule is the parsimonious feedback specification under the reported diagnostics.

7.2. When workload correction can bridge planning updates

E2 illustrates a narrower role for within-window correction. Under medium and severe errors, the selected vehicle-level fixed pair induces realized gate output above realized yard completions, and maintaining that instruction allows unresolved workload to accumulate before a revised model is available. Fast-plus-tracker feedback responds to the observed imbalance in the original stress path. In the separate aggregate reconstruction, both feedback specifications reduce the maintained $(24, 6)$ reference's late-horizon drift from 5.526 to values within 0.001 trucks per window of zero. The quota-recovery branches triggered on days one and three have post-warm-up diagnostic costs of 0.078 and 0.094, whereas the branch triggered at the day-seven evaluation boundary has cost 5.921. Because the earlier branches receive warm-up recovery time, these are trajectory-persistence results, not from-start cost estimates for three otherwise matched response delays. They describe a documented quota-recovery rule, not periodic parameter re-identification or adaptive PTO.

The tandem model adds an organizational implication. Appointment governance should connect gate access with yard workload, equipment status, exception handling, and downstream completions. Equation (2) establishes the flow linkage, but the current experiments do not isolate the marginal effect of joint authority because a gate-only and joint policy were not compared under one archived architecture.

7.3. Information, access, and deployment requirements

The archived feedback policy has a narrow input requirement: the workload correction uses the gate queue and the yard queue plus busy yard units, while the mean of up to six preceding request counts enters only its access floor. The previous quota enters the realized switching cost observed by the optional tracker, but current requests, outstanding cohorts, recent arrivals, gate-busy count, and current disruptions are not direct workload-correction inputs. A broader deployment may monitor those additional quantities for forecasting, audit, and model maintenance. Operational value also depends on authority: if the appointment book is already closed or yard capacity cannot change at the decision frequency, workload signals can diagnose imbalance but cannot implement the proposed correction.

Waiting must be evaluated together with appointment access. E3 reports lower normalized cost for fixed-pair PTO and higher confirmation for fast-plus-tracker feedback under the tested vectors. Because W_1 and W_2 change a combined congestion-and-access score, they do not identify the standalone causal effect or monetary value of unconfirmed demand. The separate post hoc threshold leaves that external valuation to the decision maker. A deployment should therefore report waiting, throughput, confirmation, unconfirmed demand, resource use, and distributional effects jointly.

8. Conclusion

This paper develops a gate–yard framework in which a terminal confirms new appointment requests and activates yard capacity while previously confirmed cohorts remain immutable. The decision clock, appointment pipeline, service state, and safe execution maps distinguish confirmation from cancellation and planning instructions from executed actions.

The computational evidence supports a conditional operating sequence. Explore-then-select fixed-pair planning is the strongest cost and waiting benchmark in E1, and a separate high-fidelity Monte Carlo-selected pair is strongest in E2. A materially overstated service model can create persistent congestion when it induces a maintained instruction whose realized gate output exceeds realized yard completions. The documented independent aggregate reconstruction shows that trajectories with earlier quota-rule activation enter the post-warm-up evaluation period with much less persistent congestion, but it is not a common-state delay experiment and neither estimates capacity nor re-solves PTO. In its feedback comparison, fast-only correction has lower aggregate diagnostic cost, terminal backlog, and late-horizon drift than fast-plus-tracker feedback, with pointwise seed-block intervals that exclude zero. The archived vehicle-level E2 design contains only fast-plus-tracker feedback, so the aggregate result is not a matched fast-only E2 ablation.

For practice, the resulting interpretation is to plan predictively, monitor both workload and model fidelity, re-estimate and re-solve when those capabilities are available, and use workload correction to bridge replanning latency. The aggregate quota-recovery rule is an illustrative response-timing diagnostic rather than a proposed substitute for adaptive PTO. Any implementation must preserve appointment access, coordinate gate and yard information, and report service and access outcomes jointly.

The principal limitations concern evidence scope rather than loss of the vehicle-level implementation. Aggregate public records discipline the scenarios but do not provide linked vehicle-level appointment–arrival–gate–yard trajectories or identify all terminal-specific parameters. E1–E3 and the high-fidelity component diagnostic are reproducible from the archived source, configurations, deterministic seed rules, execution mappings, and raw outputs. E2 nevertheless contains no matched fast-only arm and does not test alternative feedback initializers.

Tables 10 and 12 are generated by separate documented aggregate reconstructions. Their code, configurations, seeds, and path-level outputs are archived, but their absolute values are not pooled with vehicle-level E2. The seed-block intervals in Table 10 cover Monte Carlo and starting-seed variability within that reconstruction, not structural, calibration, parameter, or field uncertainty. Future work should compare fast-only and fast-plus-tracker feedback with adaptive time-varying PTO in one vehicle-level pipeline under bidirectional errors, align response triggers at a common evaluation state, and add a controlled gate-only-versus-joint-authority ablation.

Data and code availability

A reproduction archive aligned with this submission accompanies the manuscript. It contains the Python source for the vehicle-level gate–yard simulator, experiment configurations, deterministic seed rules, cost and execution mappings, fixed-pair PTO candidate searches, processed paper-date calibration inputs, raw replication outputs, tests, and table-generation scripts. Its version record identifies the matching manuscript source and PDF by SHA-256 hash. Archive verification reproduces E1, E2, E3, and the high-fidelity component diagnostic exactly; the E2 archive also retains all 380 candidate-search records.

For E0, the archive contains the processed paper-date inputs, source and transformation records, download and replay scripts, hashes, and verification outputs. An author-retained third-party raw snapshot dated 2026-07-09 reproduces all eight archived E0 outputs and all 24 scenario parameters exactly. That raw snapshot is not redistributed in the supplied archive, so the paper-date raw-to-processed chain cannot be replayed from the supplied materials alone. Live public dashboards are date-sensitive, and future downloads are not expected to match the paper-date bytes; third-party files remain subject to their providers' access and licensing terms.

The archive also contains the code, configurations, seeds, path-level results, and seed-block summaries for the documented independent aggregate reconstructions reported in Tables 10 and 12. Earlier scripts for these two diagnostics were not retained. The reported tables are generated from the current documented reconstructions and are not pooled numerically with the vehicle-level E2 results.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the authors used OpenAI Codex to review manuscript organization, improve language clarity, and assist with LaTeX editing. The authors reviewed and edited all resulting material and take full responsibility for the content of the publication.

Appendix A. Proofs of the analytical results

Proof of Proposition 1

Continuous differentiability of $\overline{F}^{+}(s^{\star}, \cdot)$ and $\det J_F(\theta^{\star}) \neq 0$ satisfy the inverse function theorem. The relaxed next-window response map consequently has a continuously differentiable local inverse around $\theta^{\star}$, establishing local uniqueness and continuous dependence on the target feedback vector. □

Proof of Proposition 2

For either counterfactual action, Eq. (2) gives $N_{t+1}^{y,\pi} - N_t^{y} = D_t^{g,\pi} - D_t^{y,\pi}$. Taking conditional expectations from the same full state ξ under the same conditional exogenous law and applying $\Delta_t^{\mathrm{FB}}(\xi) \leq \Delta_t^{\mathrm{PTO}}(\xi) - \gamma$ yields Eq. (25). □

Proof of Proposition 3

PTO optimality under the belief model gives $\widehat{J}(u^{\mathrm{PTO}}) \leq \widehat{J}(u)$ for every $u \in \mathcal{U}$. When $\widehat{J}(u) = J(u)$ throughout this open-loop class, substitution yields Eq. (27). □

Proof of Proposition 4

The uniform model-error bound and PTO optimality give

$$J(u^{\mathrm{PTO}}) \leq \widehat{J}(u^{\mathrm{PTO}}) + \Delta_{\mathcal{U}} \leq \widehat{J}(u^{*}) + \Delta_{\mathcal{U}} \leq J(u^{*}) + 2\Delta_{\mathcal{U}},$$

which is Eq. (28). For the finite-sample extension, on the stated uniform-error event,

$$\begin{aligned} J(u_N^{\mathrm{MC}}) &\leq \widehat{J}(u_N^{\mathrm{MC}}) + \Delta_{\mathcal{U}} \\ &\leq \widetilde{J}_N(u_N^{\mathrm{MC}}) + \Delta_{\mathcal{U}} + \varepsilon_N \\ &\leq \widetilde{J}_N(u^{*}) + \Delta_{\mathcal{U}} + \varepsilon_N \\ &\leq \widehat{J}(u^{*}) + \Delta_{\mathcal{U}} + 2\varepsilon_N \\ &\leq J(u^{*}) + 2\Delta_{\mathcal{U}} + 2\varepsilon_N. \end{aligned}$$

This is Eq. (28a). □

Proof of Proposition 5

Equation (2) gives $N_{t+1}^{y} - N_t^{y} = D_t^{g,\mathrm{PTO}} - D_t^{y,\mathrm{PTO}}$. Taking expectations and applying Eq. (29) yields Eq. (30); summing the same identity over K windows yields Eq. (31). In addition, $D_t^{y,\mathrm{PTO}} \leq Y_t(\mu_t^{y,\mathrm{PTO}})$, so the potential-capacity condition stated after Proposition 5 is sufficient for Eq. (29). □

Execution-map feasibility argument

By construction, Γ_t^{FB} applies the physical quota and capacity bounds, the pre-specified integer mapping, and the compatible access floor. Whenever $\Theta_t^{\mathrm{FB},d}$ is nonempty, its output therefore belongs to $\Theta_t^{\mathrm{FB},d}$, which proves Eq. (32). This argument establishes membership only and makes no nearest-point claim. □

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